

The abdominal wall

Department of Anatomy

The abdomen (1)

Second Part

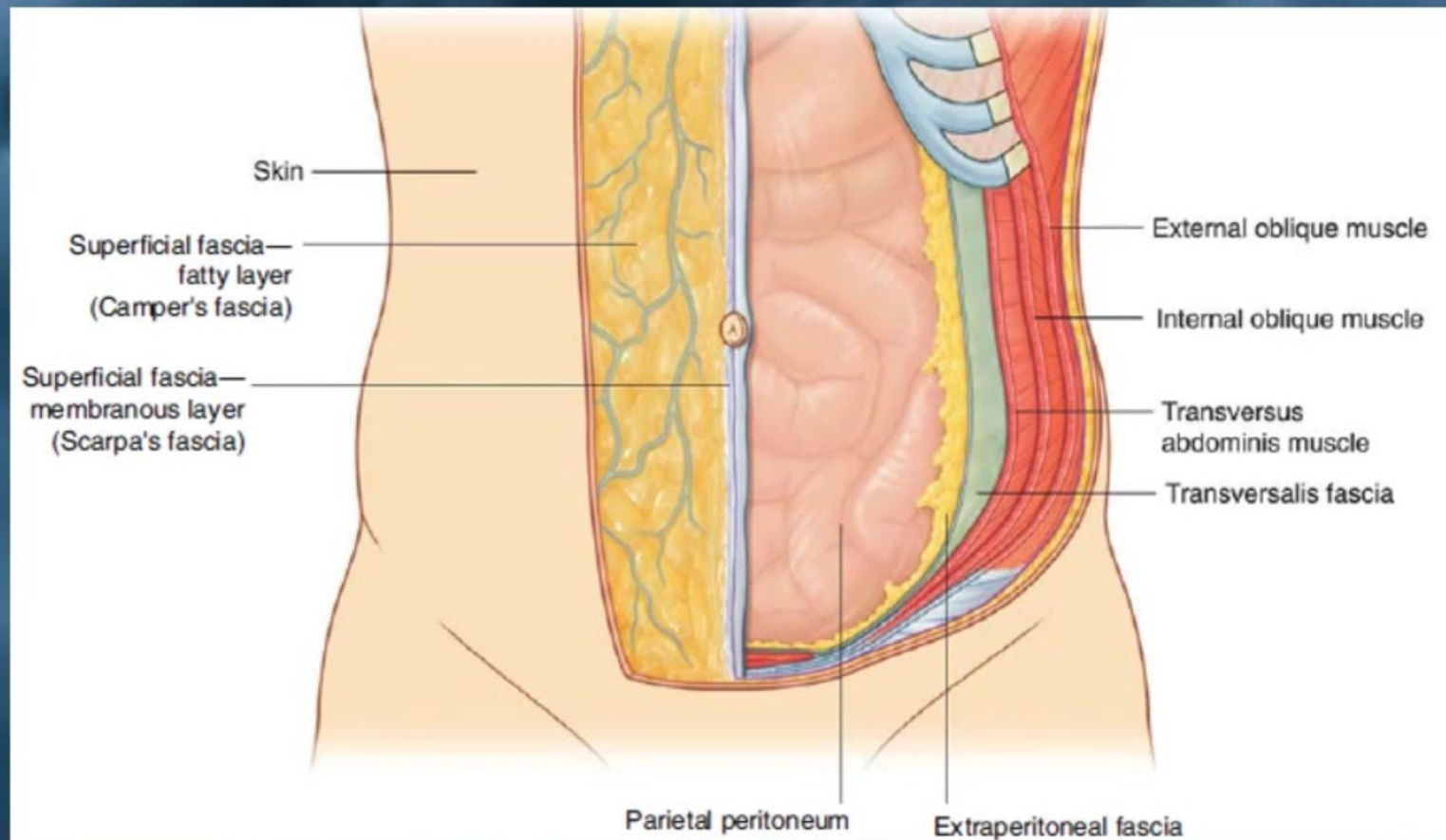
Dr. Osman Ahmed Osman

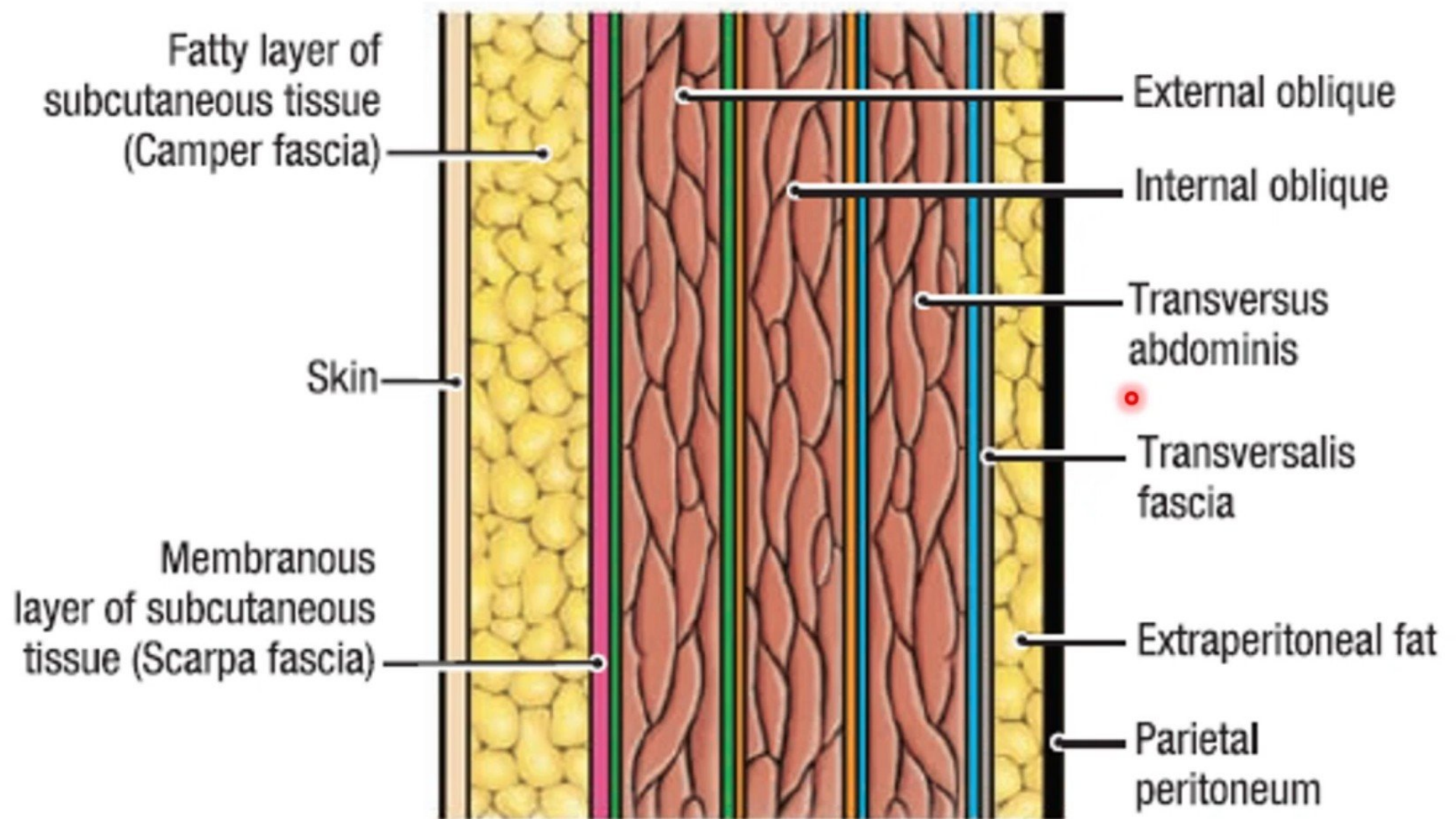
Lecture points

- Location of Abdomen.
- The abdomen region in anatomical base
- The superior aperture of the abdomen.
- The component parts of abdominal walls.
- Pelvic Inlet.
- Relationship of the regions to the abdomen.
- The abdominal cavity is continuous with the pelvic cavity.
- The component parts of abdominal walls.
- Abdominal Cavity
- The abdominal Viscera
- The peritoneal cavity and peritoneum
- Arterial supply of abdominal organs
- Abdomen assist in breathing
- Posterior Abdominal Wall
- Boundaries of the Abdomen
- Anatomical landmark
- The division of abdomen into four quadrants
- The nine region of the abdomen.
- The Anterior Abdominal Wall Functions.
- **The Layers of anterior abdominal wall.**
- Transvers section showing the layers of the abdominal wall.
- The Superficial fascia.
- Associated ligaments.
- Muscles of the anterior abdominal wall.
- The lumbar triangle.
- Rectus sheath.
- The inguinal ligament.
- Cutaneous innervation Of the anterior abdominal wall.
- Blood vessels of the anterior abdominal wall
- Venous Drainage.
- Lymphatic Drainage

The Layers of anterior abdominal wall

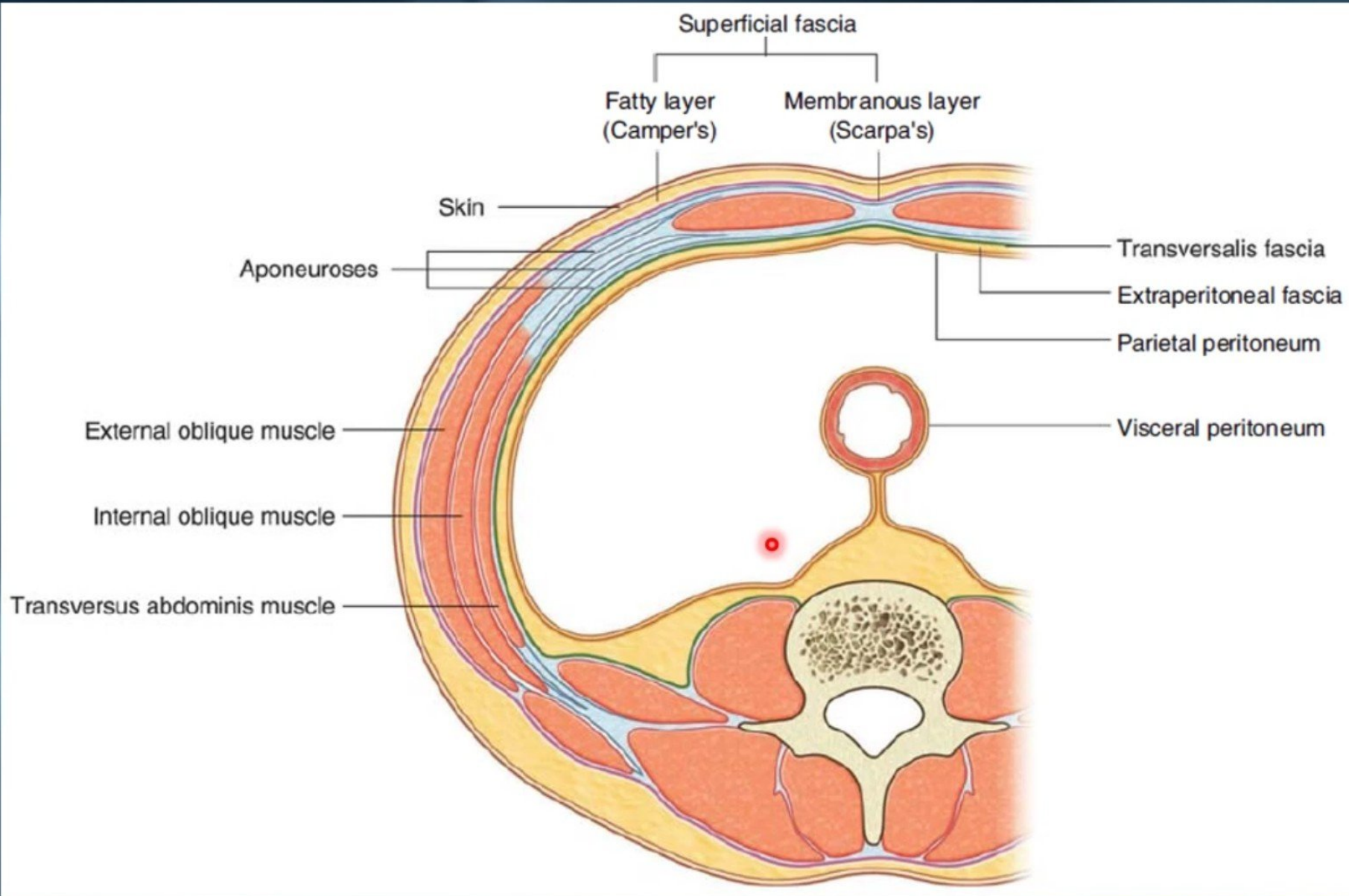
- Skin
- Superficial fascia (two layers).
 - superficial fatty layer**(CAMPER'S fascia).
 - deep membranous layer**(SCARPA'S fascia).
- No deep fascia.
- External oblique muscle.
- Internal oblique muscle.
- Transversus abdominis muscle.
- Transversalis fascia.
- Lateral to midline (linea alba)- rectus abdominis muscle.
- Extraperitoneal tissue layer- peritoneum.





C. Longitudinal Section

Transvers section showing the layers of the abdominal wall

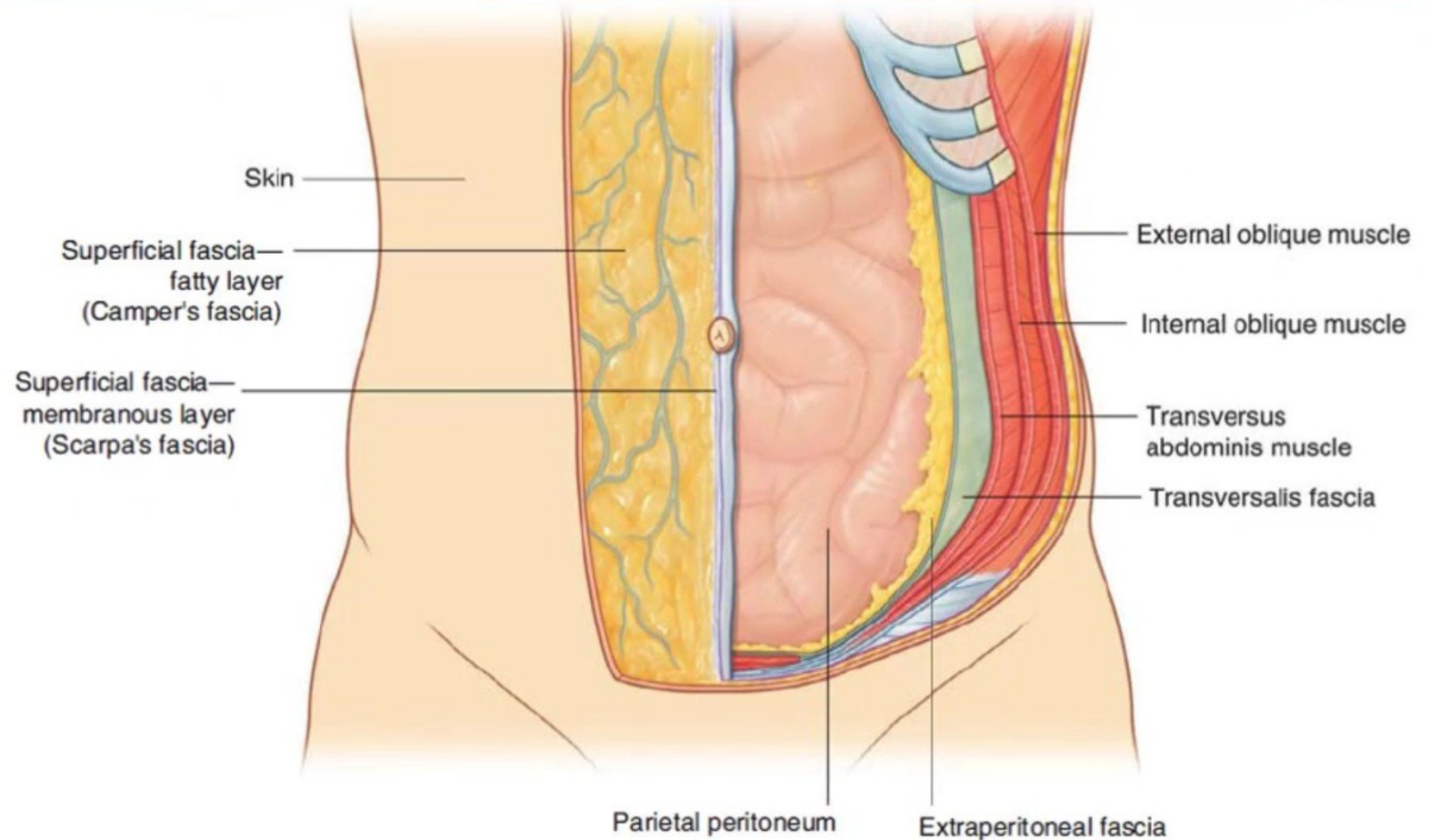


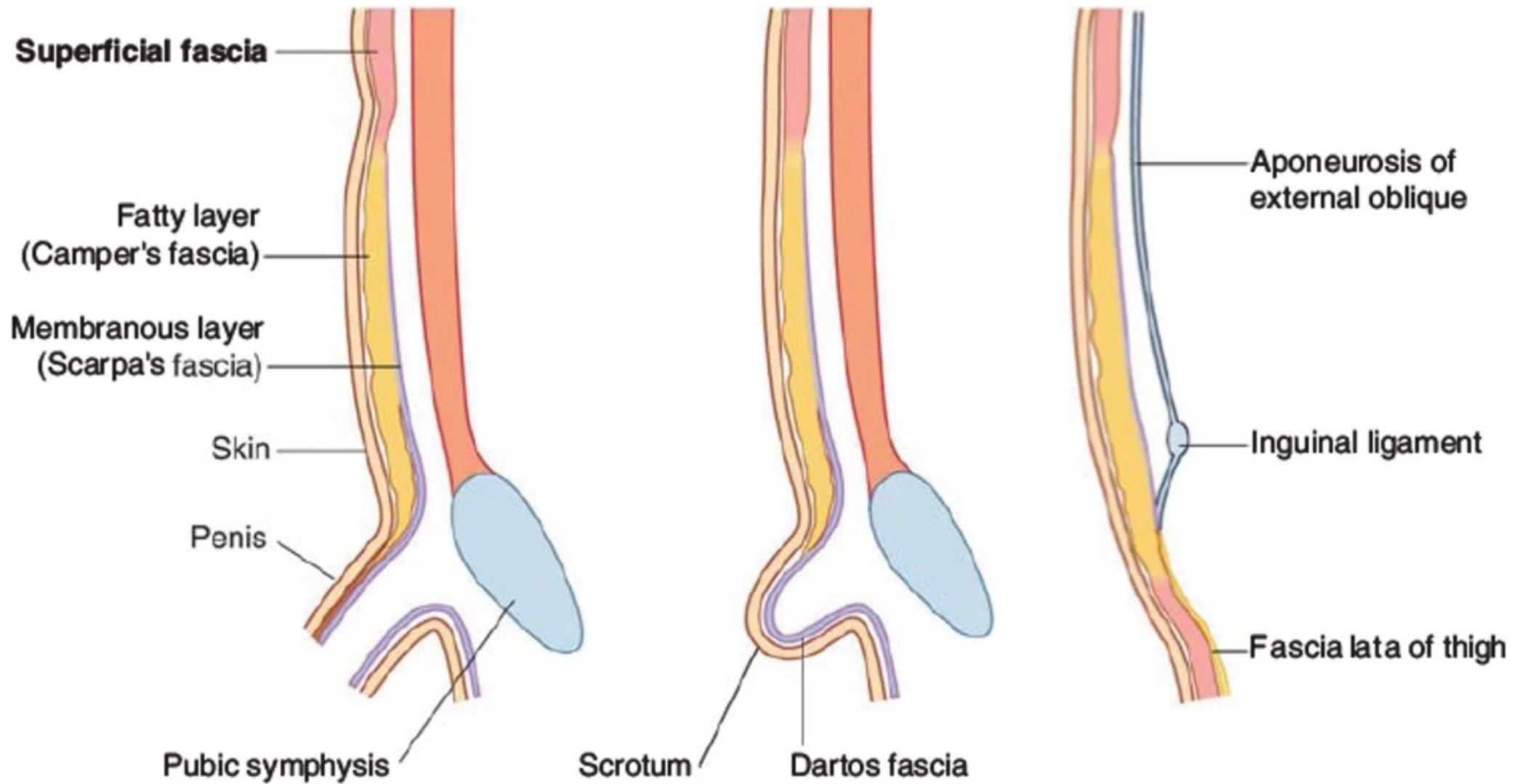
The Superficial fascia

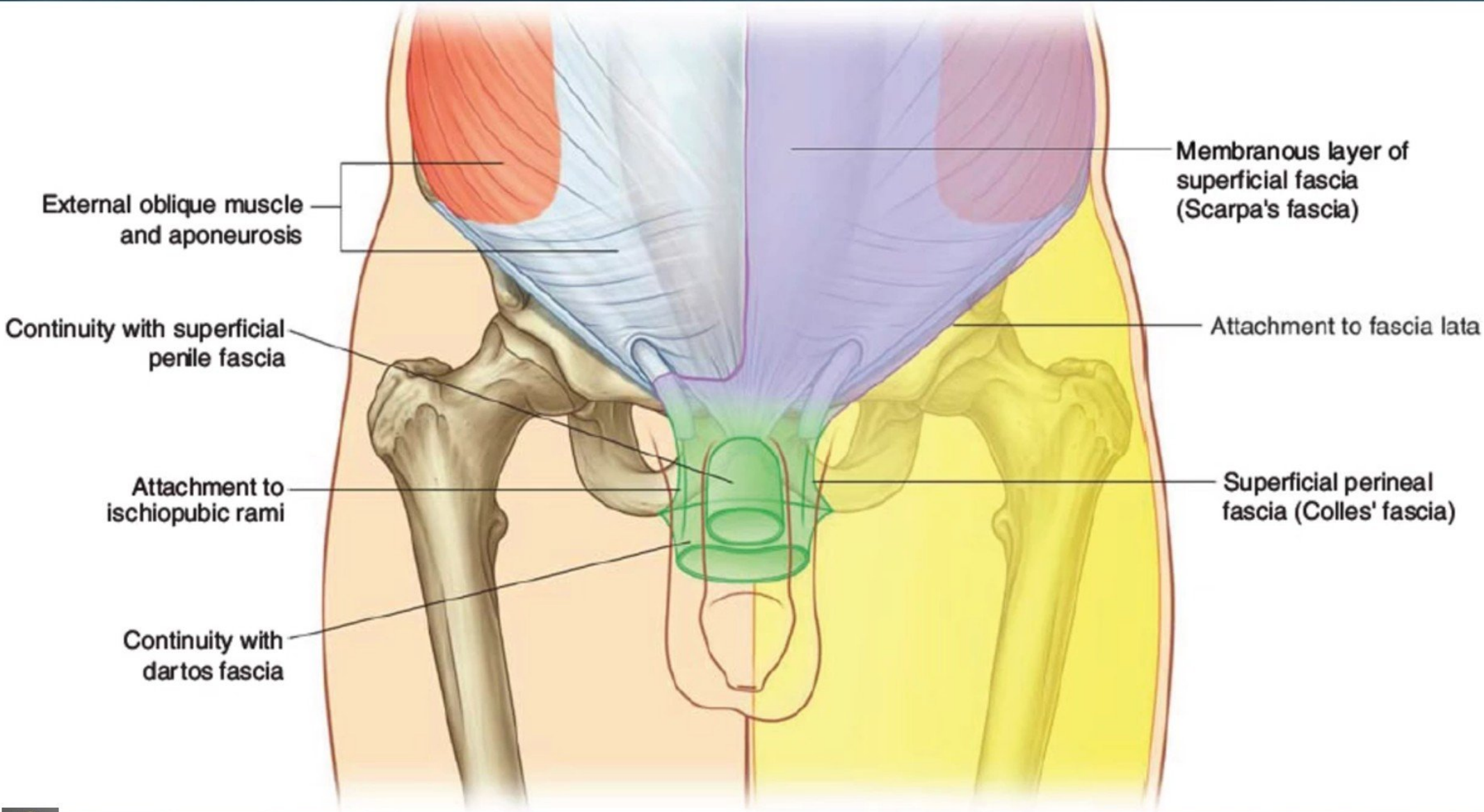
The superficial fascia of the abdominal wall (subcutaneous tissue of abdomen) is a layer of fatty connective tissue.

There are two layers:

- 1) Superficial fatty layer (CAMPER'S fascia).**
- 2) Deep membranous layer (SCARPA'S fascia).**

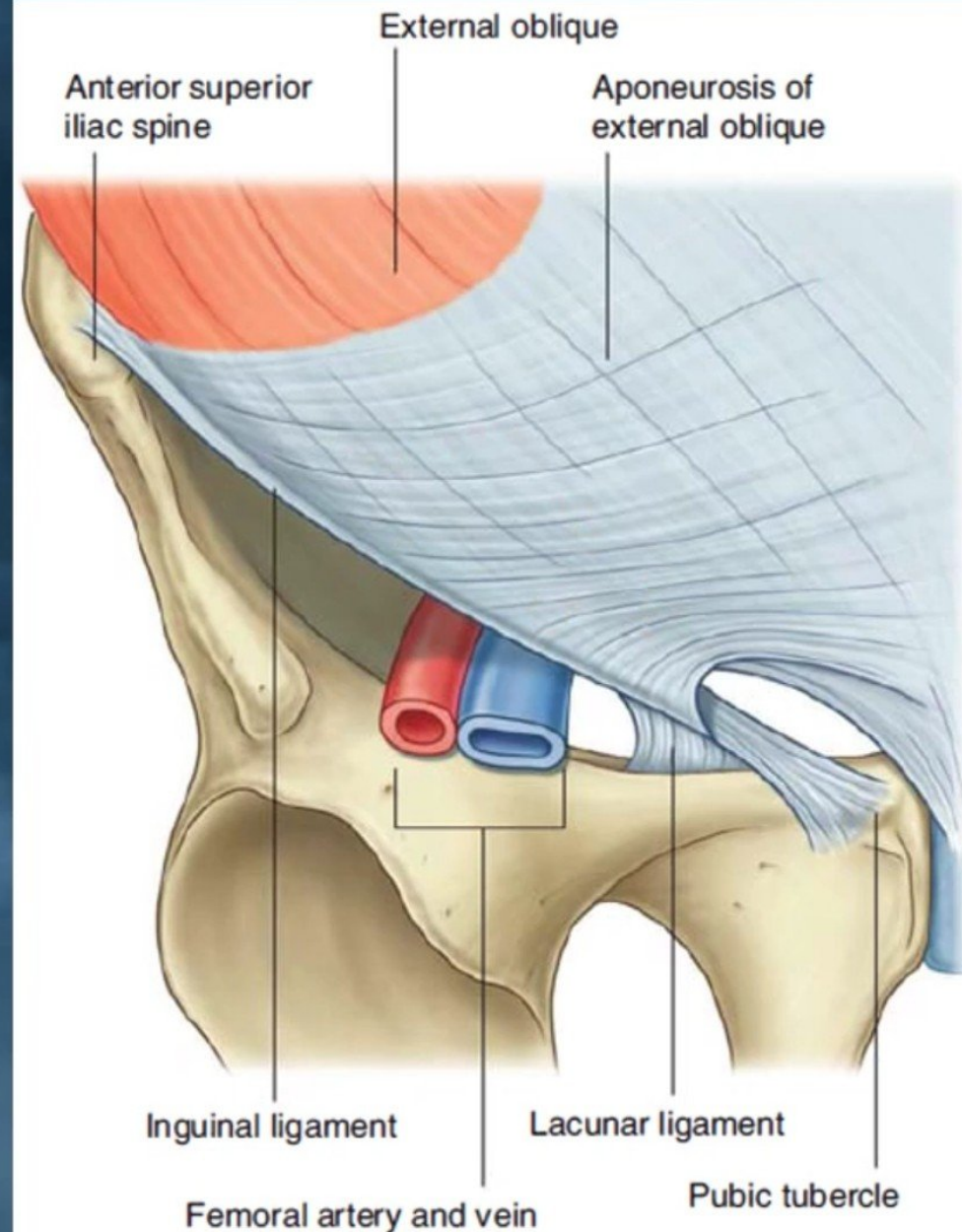






Associated ligaments

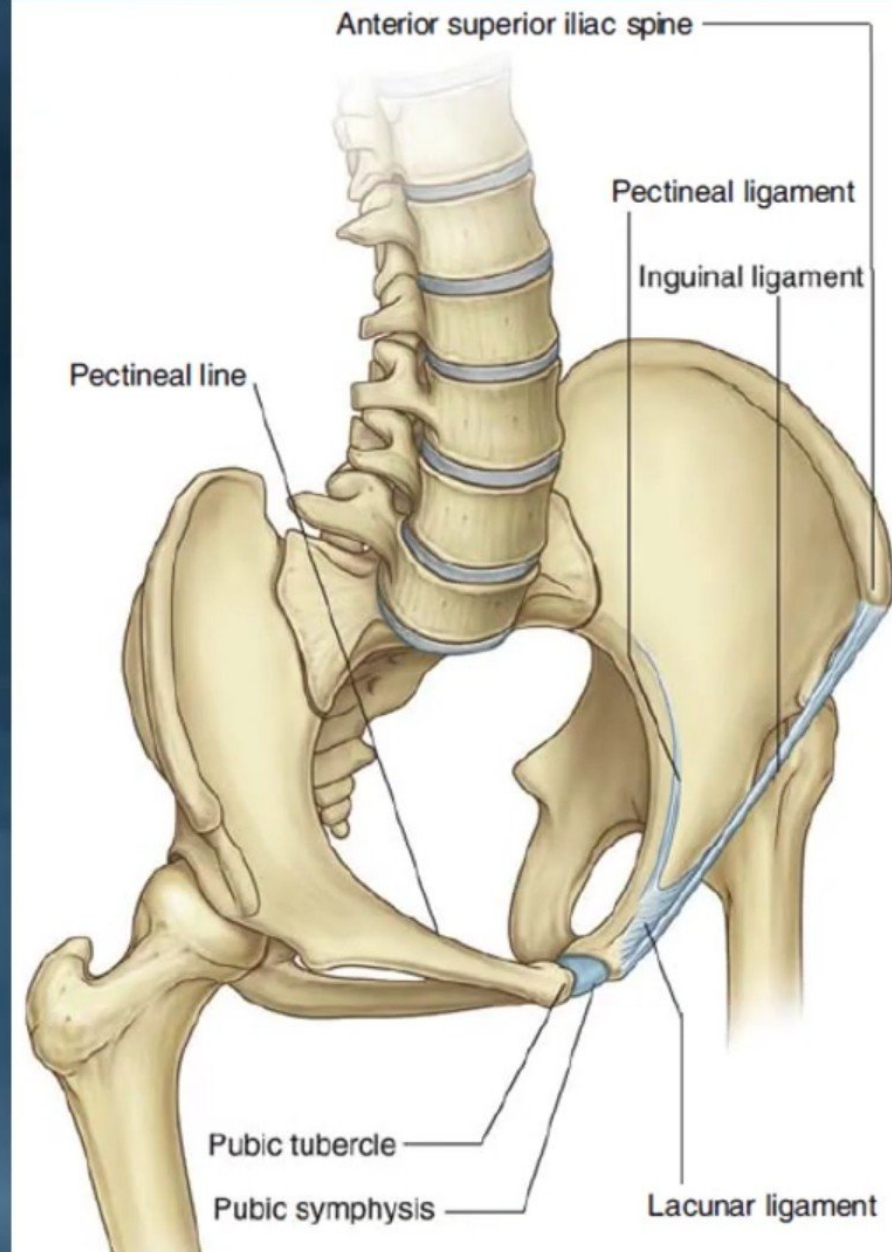
- The lower border of **external oblique aponeurosis** forms **the inguinal ligament** on each side .
- This thickened reinforced free edge of **the external oblique aponeurosis** passes between **the anterior superior iliac spine laterally** and **the pubic tubercle medially**.
- It folds under itself forming a trough, which plays an important role in the formation of **the inguinal canal**.

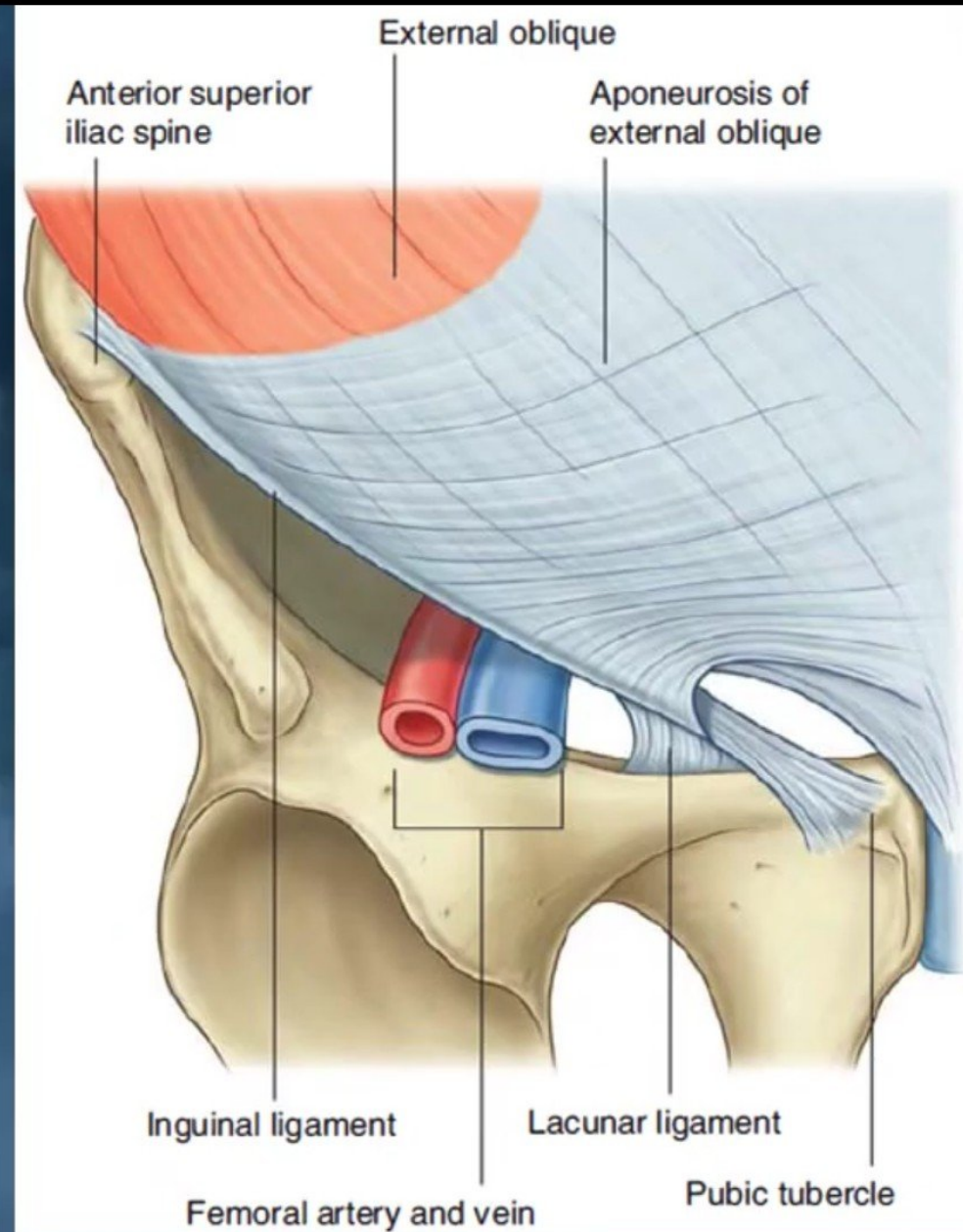
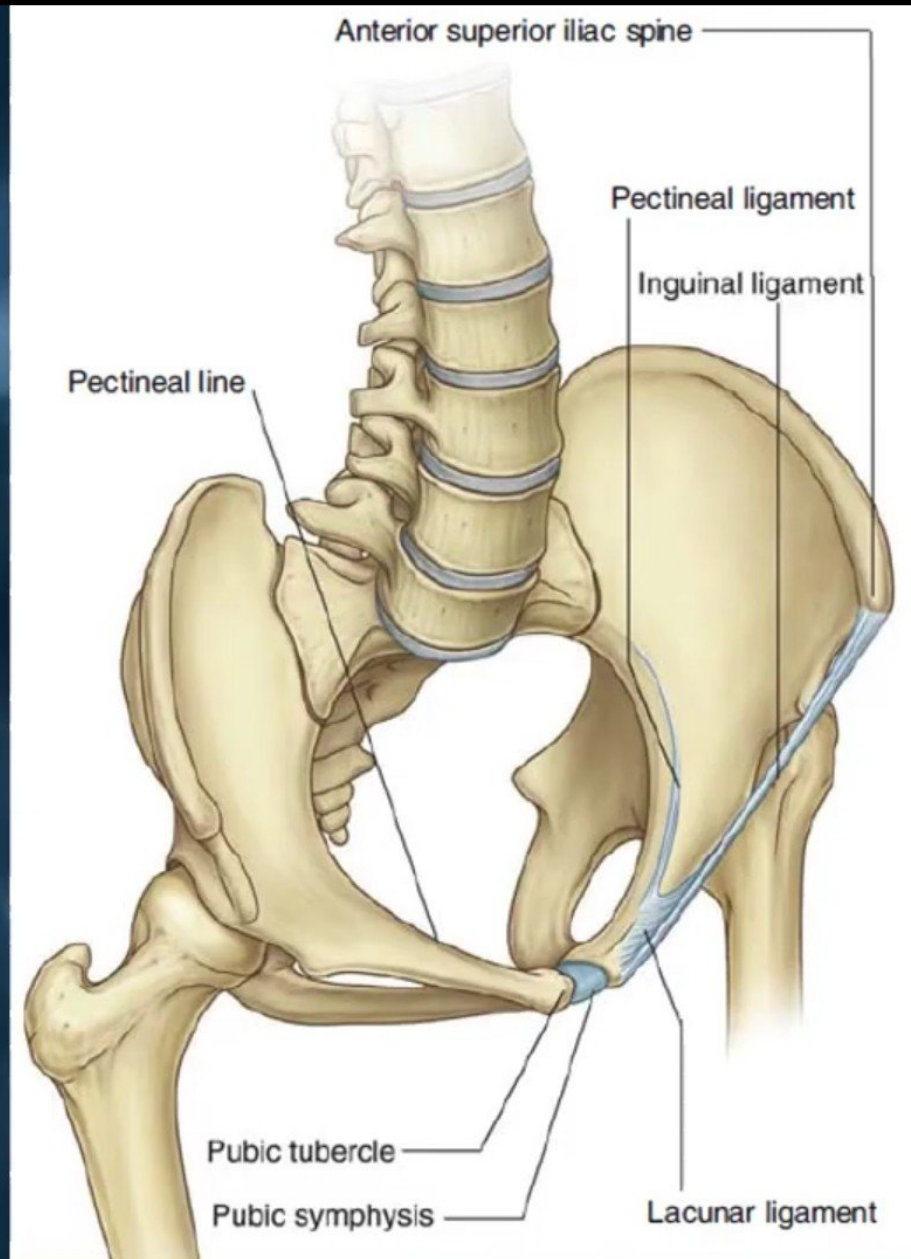


Associated ligaments

Several other ligaments are also formed from extensions of the fibers at the medial end of the inguinal ligament:

- The **lacunar ligament** is a crescent – shaped extension of fibers at the medial end of the inguinal ligament that pass backward to attach to the **pecten pubis** on the **superior ramous of pubic bone**.
- Additional fibers extend from the lacunar ligament along the pecten pubis of the pelvic brim to form the **pectineal (Cooper's) ligament**.





Muscles of the anterior abdominal wall

- **External oblique muscle** (most superficial muscles).
- **Internal oblique muscle.**
- **Transversus abdominis muscle.**
 - Those 3 muscles are laterally located.
- ✓ On both sides of the midline (linea alba)
- **Rectus abdominis muscle.**
- **Pyramidalis muscle (inferiorly).**

External oblique

Origin:

- from each of the lower eight ribs (5 – 12).
- The muscle has a free posterior border.

Insertion:

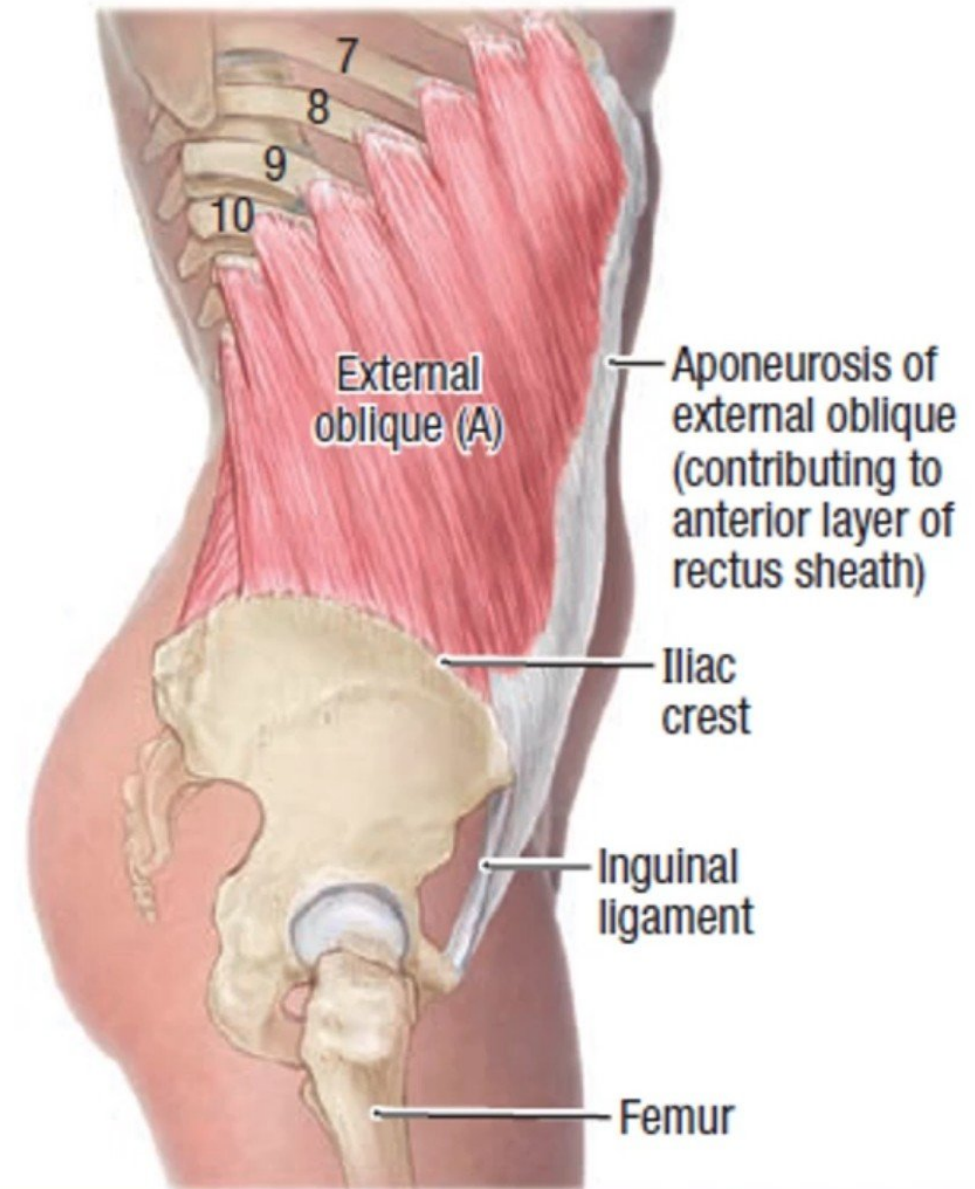
- its insertion by fleshy fibres into the anterior half of the outer lip of the iliac crest.
- The aponeurotic fibres, directed obliquely downward across the front of the rectus abdominis along the whole length of the linea alba from the fifth rib to the xiphisternum.

Nerve supply:

- Anterior rami of the lower Intercostal (T7–T11) and subcostal (T12) nerves.

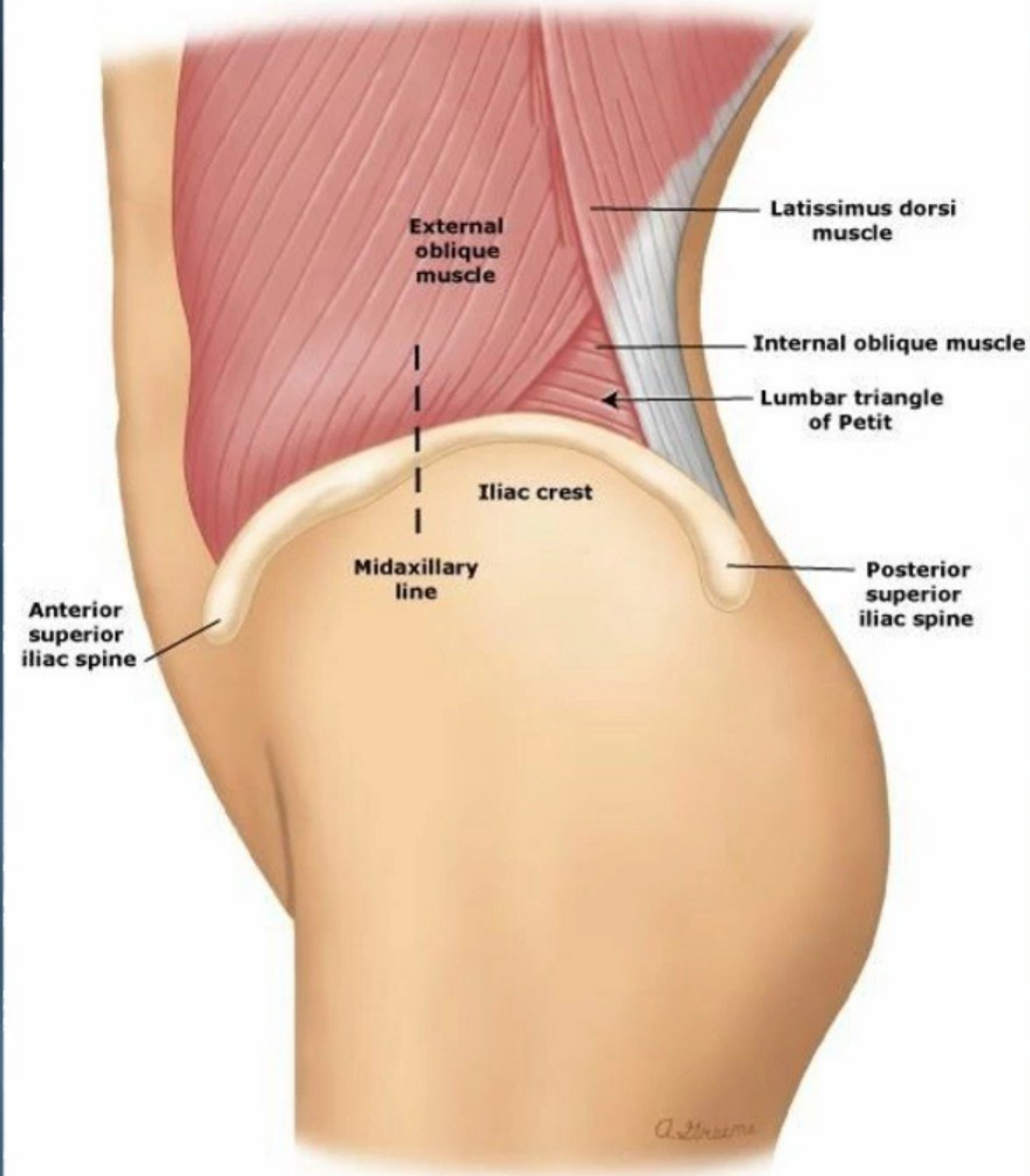
Action:

- Compresses abdomen.
- Flexes trunk.
- Active in forced expiration.



The lumbar triangle (of Petit):

- It is found in the free posterior border of the muscle
- Its floor is formed by the **internal oblique**.
- It is bounded behind by the anterior border of **latissimus dorsi** and below by **the iliac crest**.
- The triangle may be the site of a rare **lumbar hernia**.



Internal oblique

Origin:

- Lateral two-thirds of inguinal ligament.
- Iliac crest : between origin of external and transversus abdominis muscles.
- Thoracolumbar fascia.

Insertion:

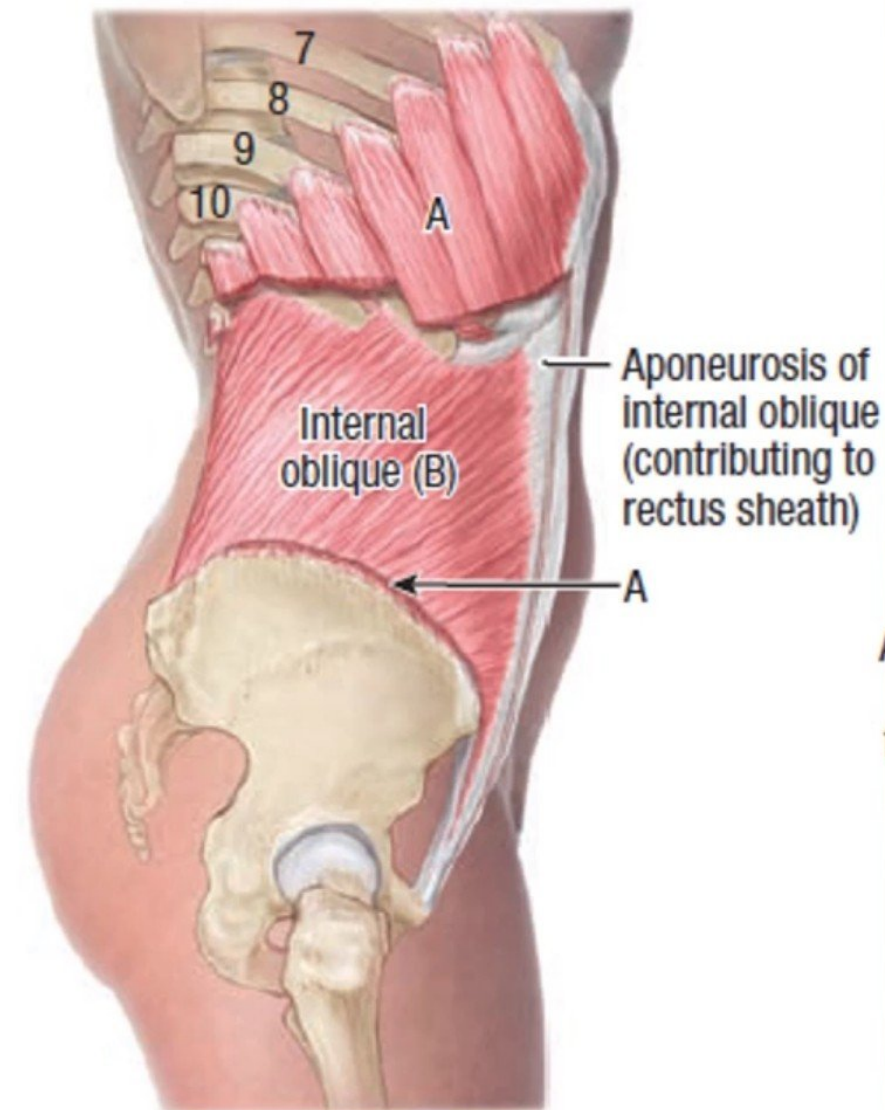
- Lower four costal cartilages;
- Linea alba.
- Pubic crest.
- Pectineal line.

Nerve supply:

- Intercostal (T7–T11).
- Subcostal (T12).
- Iliohypogastric (L1).
- ilioinguinal (L1).

Action:

- Compresses abdomen.
- Flexes trunk.
- Active in forced expiration.



B. Lateral View

Transverse abdominis muscle

Origin:

- Lateral one-third of inguinal ligament.
- iliac crest.
- Thoracolumbar fascia.
- lower six costal cartilages.

Insertion:

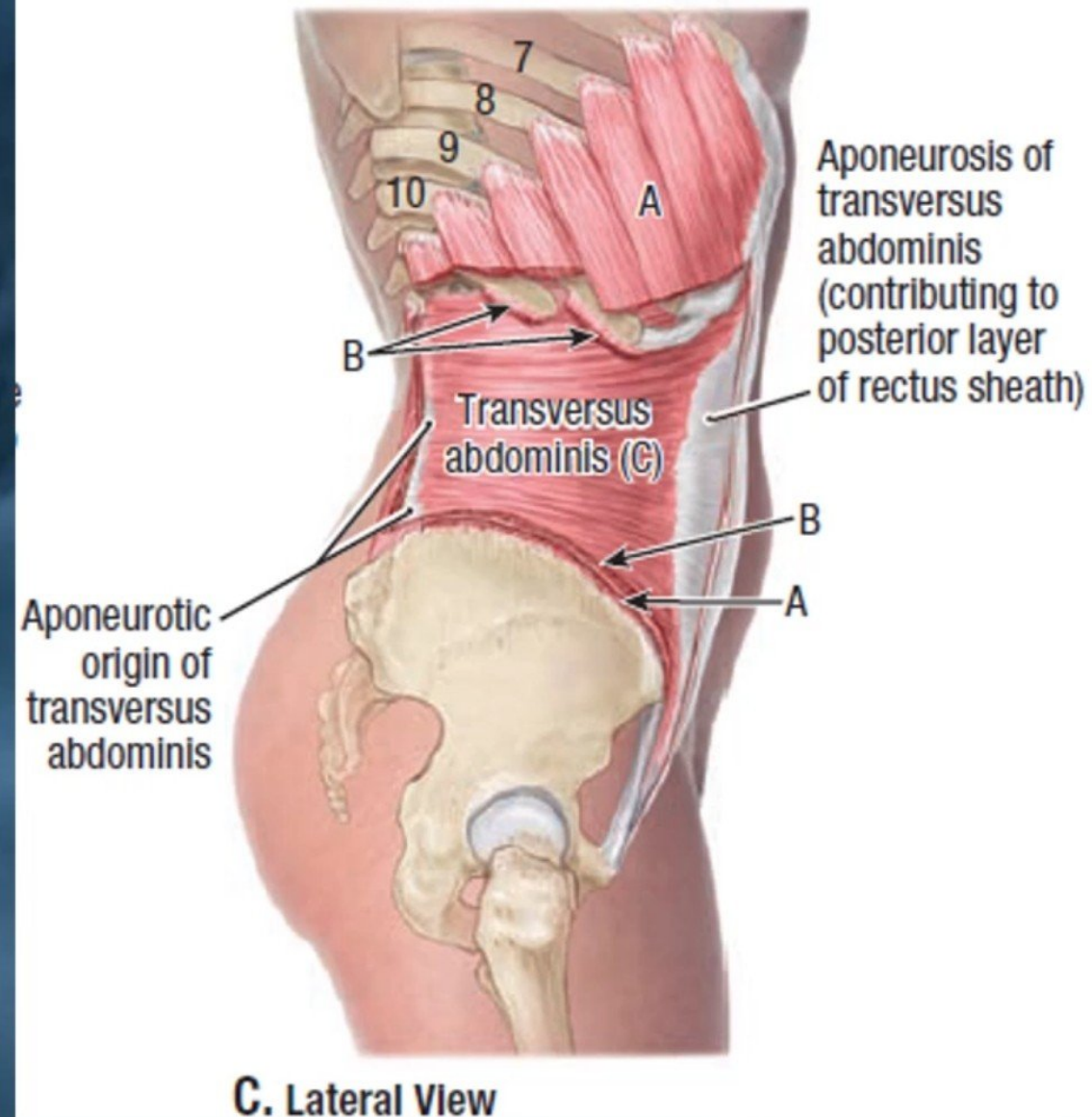
- Linea alba;
- pubic crest;
- pectineal line

Nerve supply:

- Intercostal (T7–T11).
- subcostal (T12).
- Iliohypogastric (L1).
- ilioinguinal (L1).

Action:

- Compresses abdomen.
- Flexes trunk.
- Active in forced expiration.



Rectus abdominis

Origin:

- Pubic crest.
- Pubic tubercle.
- Pubic symphysis.

Insertion:

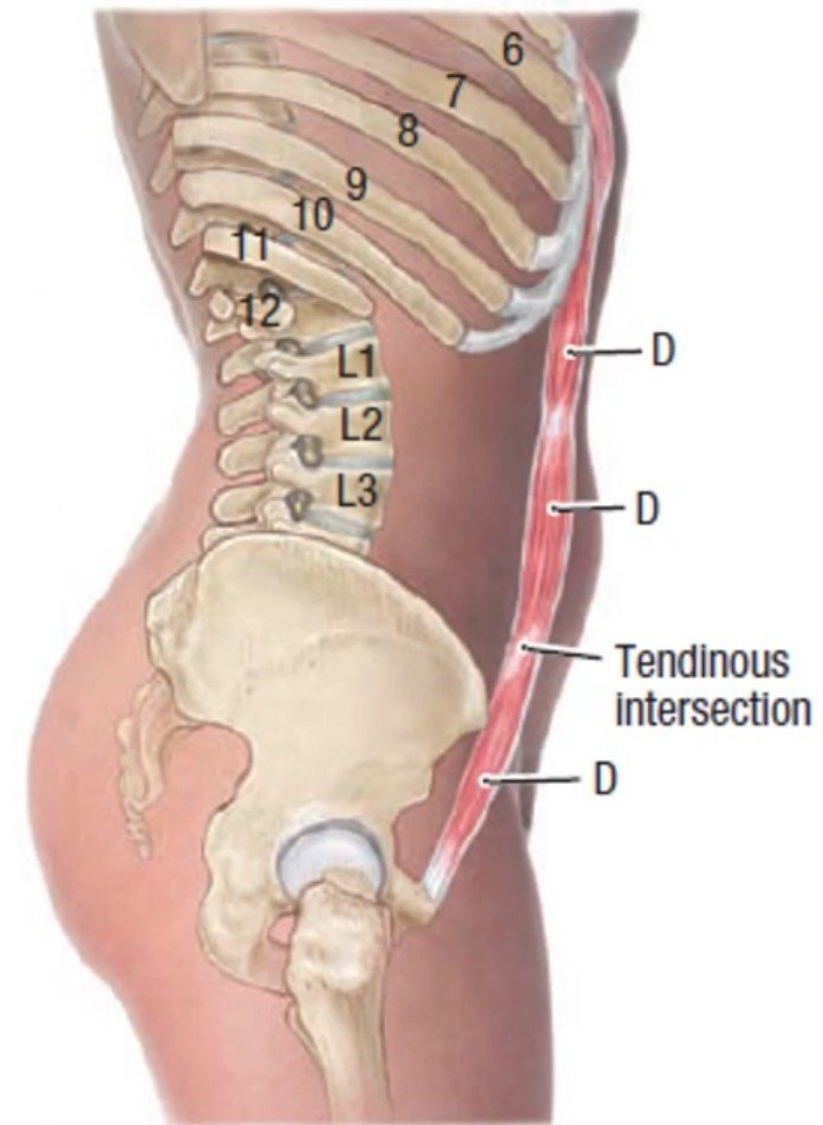
- Xiphoid process
- Costal cartilages fifth to seventh.

Nerve supply:

- Intercostal (T7–T11);
- subcostal (T12).

Action:

- Depresses ribs.
- flexes trunk



E. Lateral View

Pyramidalis muscle

The small triangular **pyramidalis** muscle found between rectus abdominis and its sheath.

Origine:

- Pubic body.

Insertion:

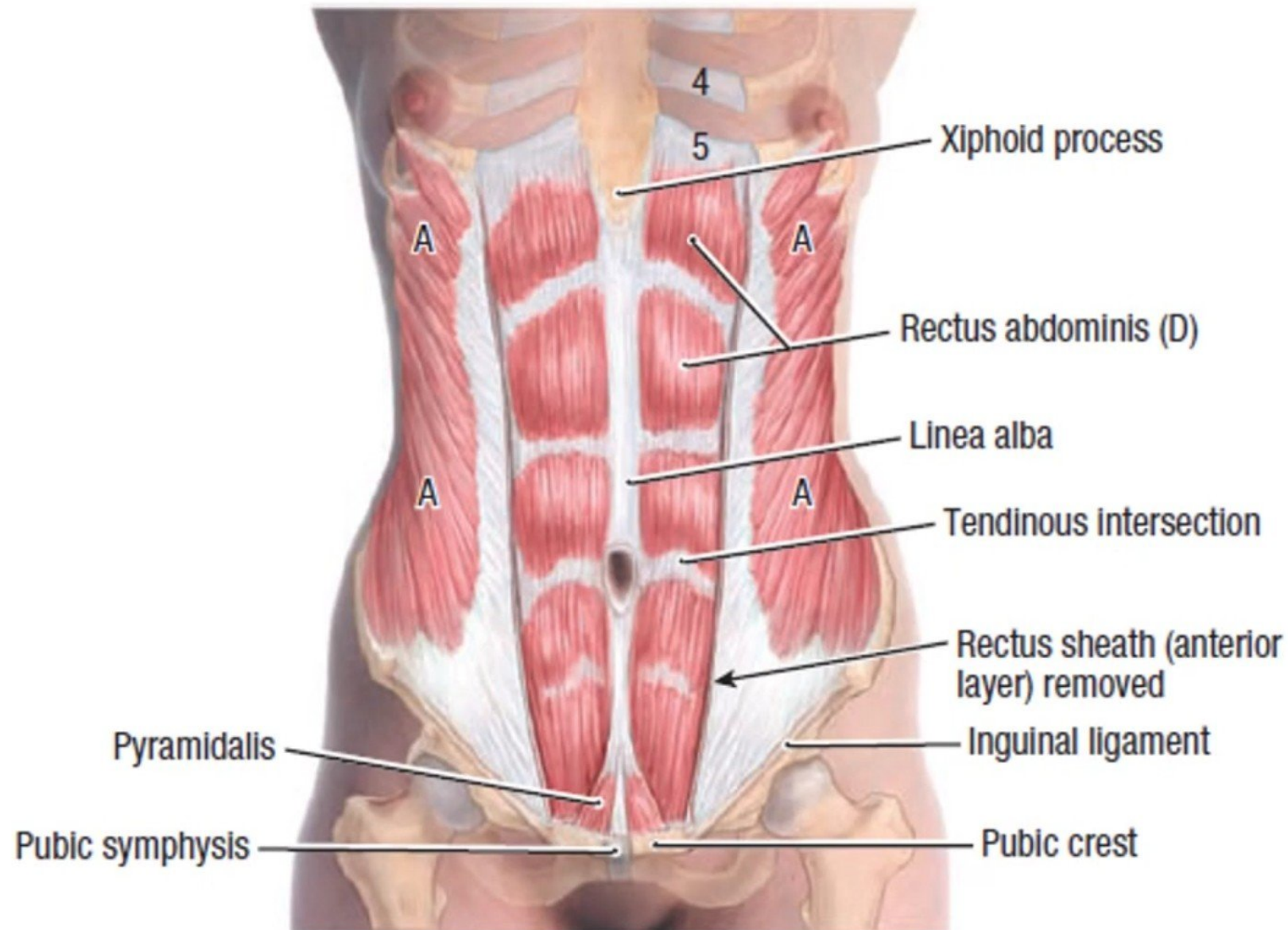
- Linea alba

Inervation:

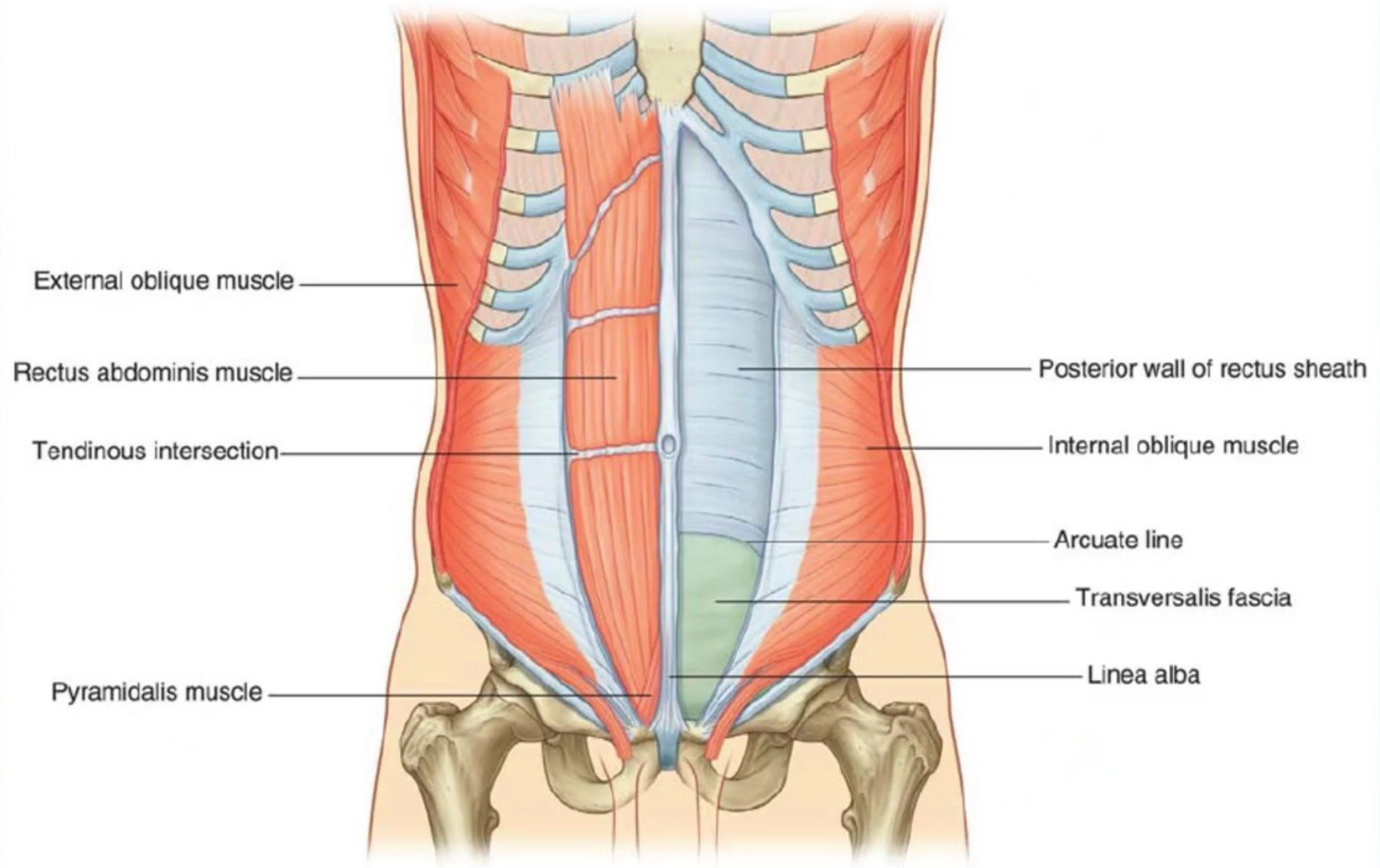
- Subcostal (T12).

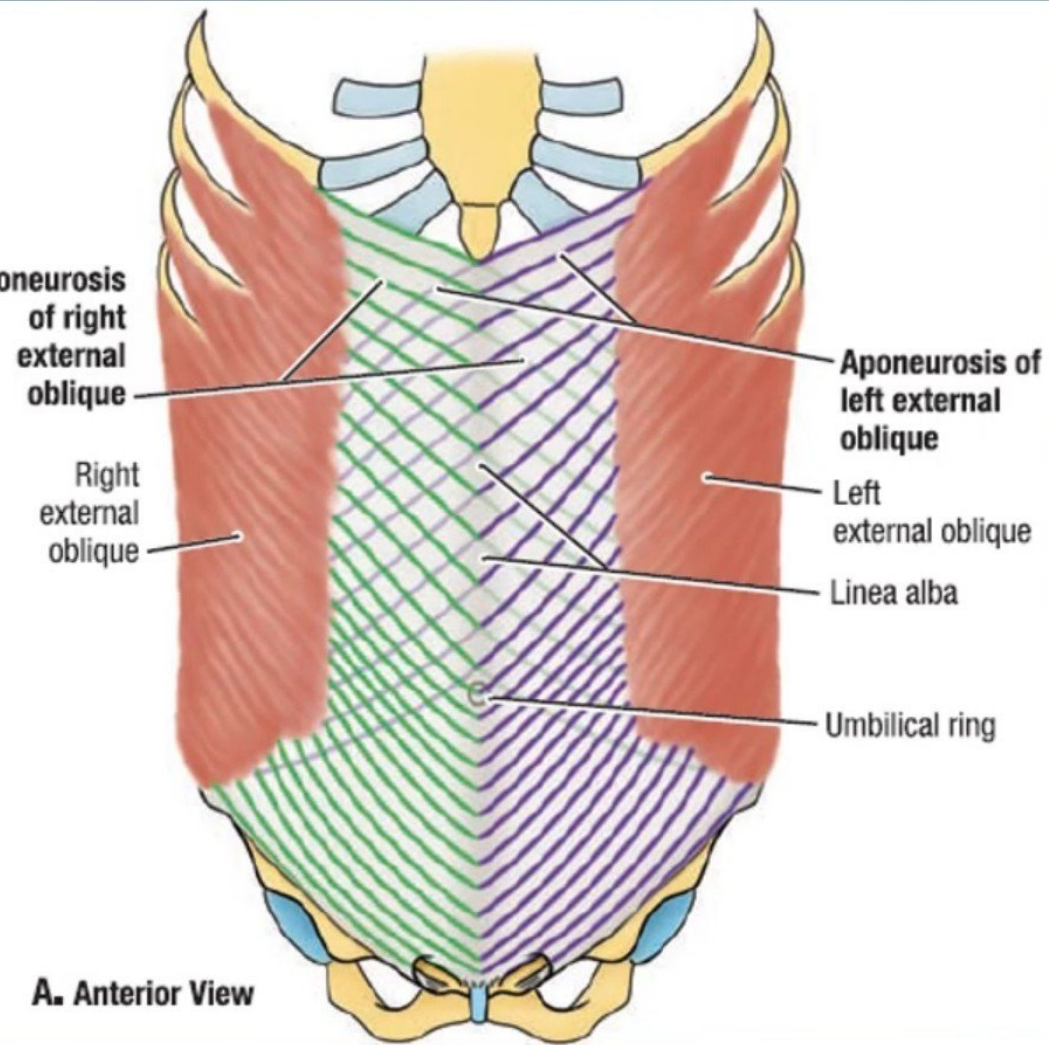
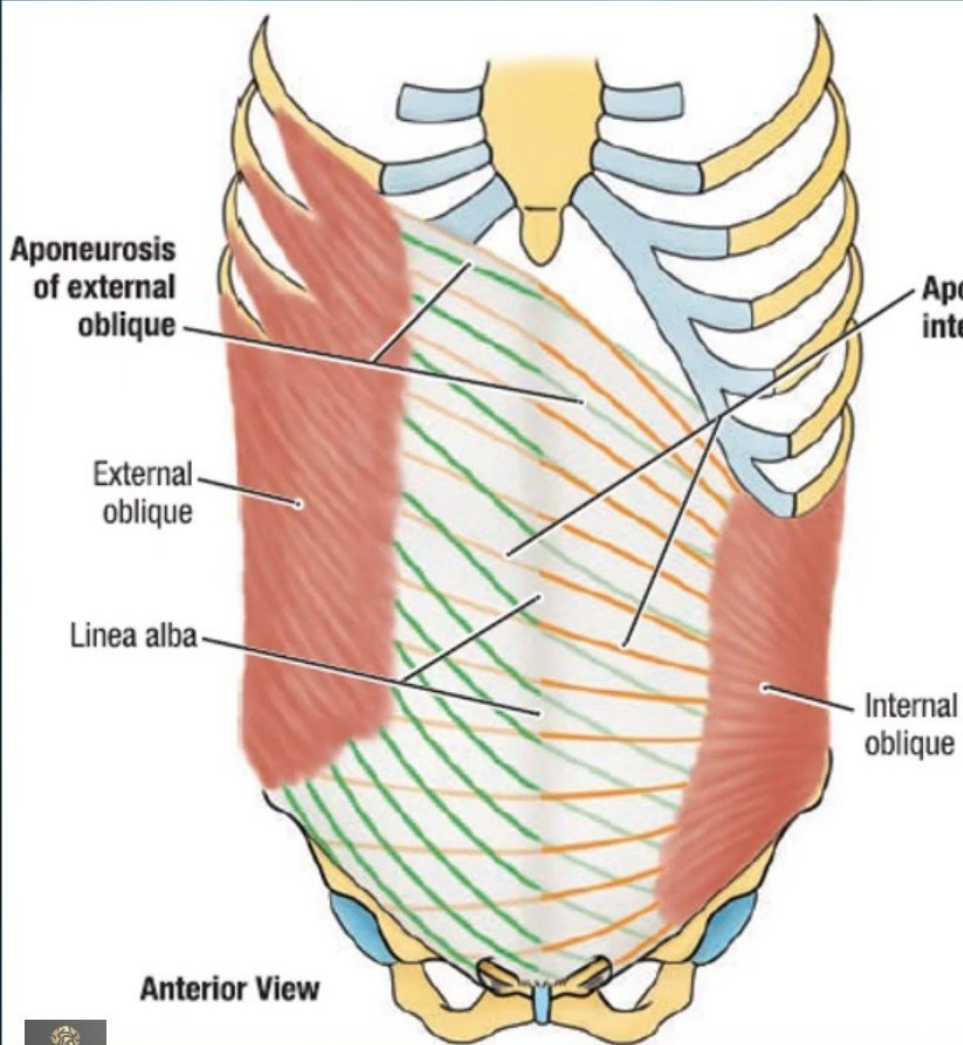
Action:

Tenses linea alba



D. Anterior View



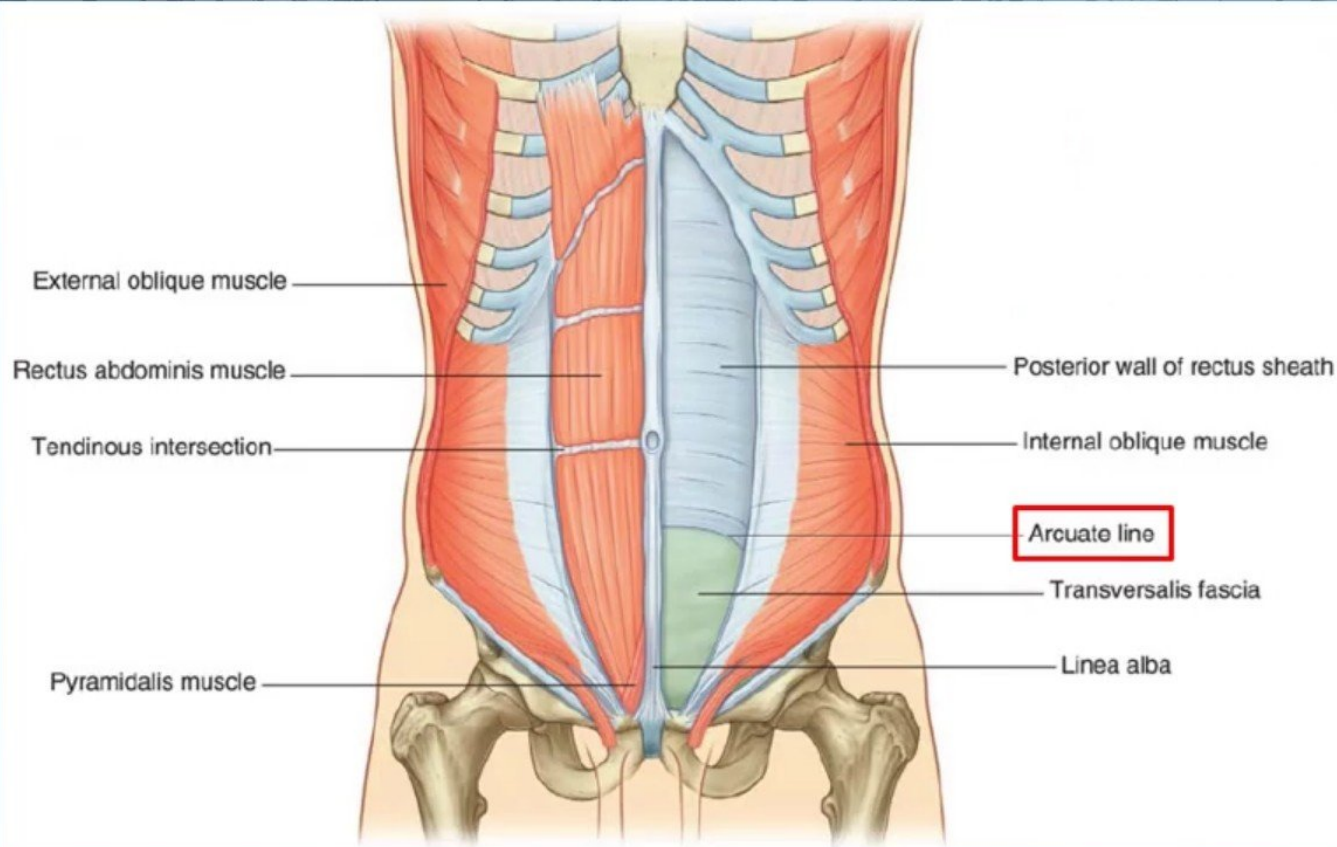
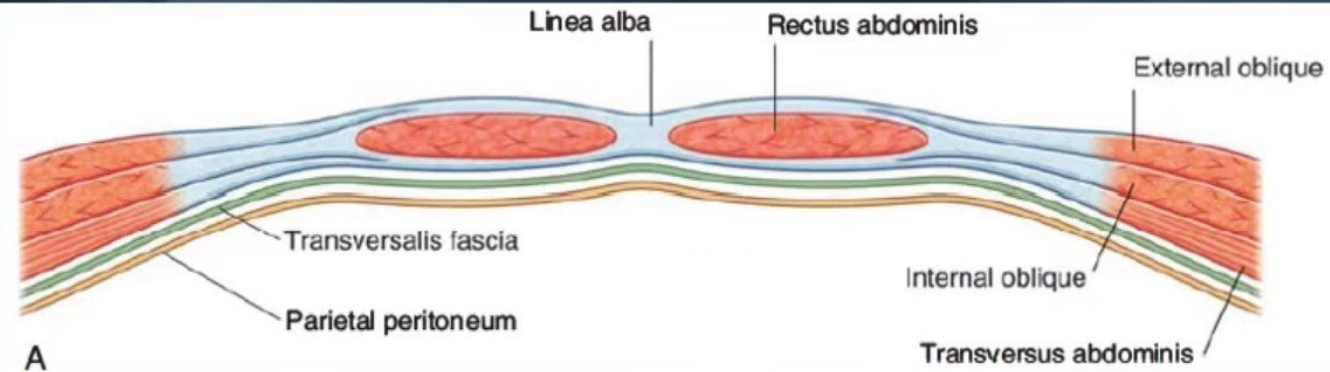


Anterior Abdominal Wall

Layer of anterior abdominal wall:

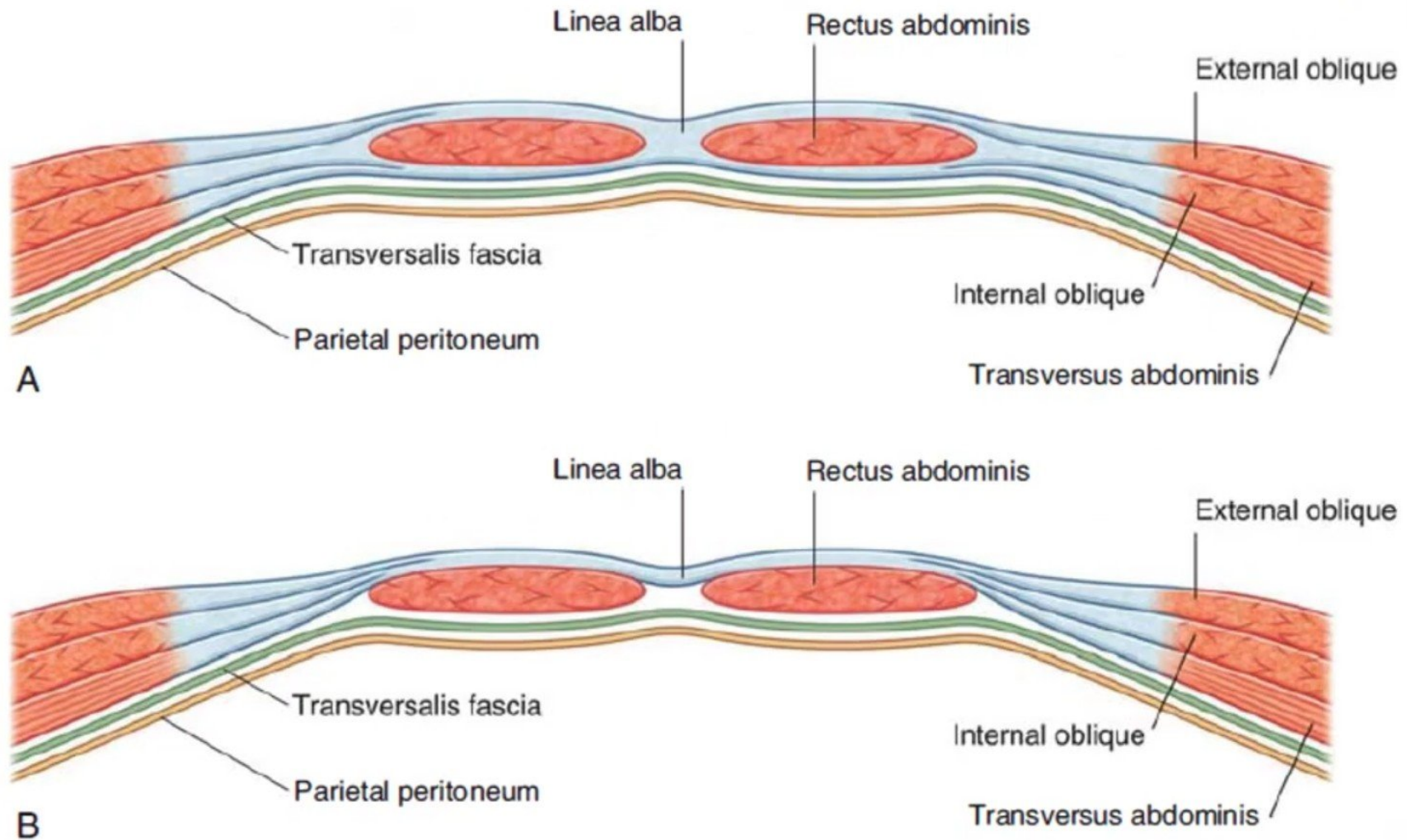
Medial:

- 1- Skin.
- 2- Superficial fascia.
- 3- External oblique muscle aponeurosis.
- 4- **Anterior lamina of the aponeurosis of the Internal oblique muscle.**
- 5- **Rectus muscle.**
- 6- **Posterior lamina of the aponeurosis of the Internal oblique muscle.**
- 7- Transversus abdominis muscle aponeurosis.
- 8- Fascia transversalis.
- 9- Peritoneum.



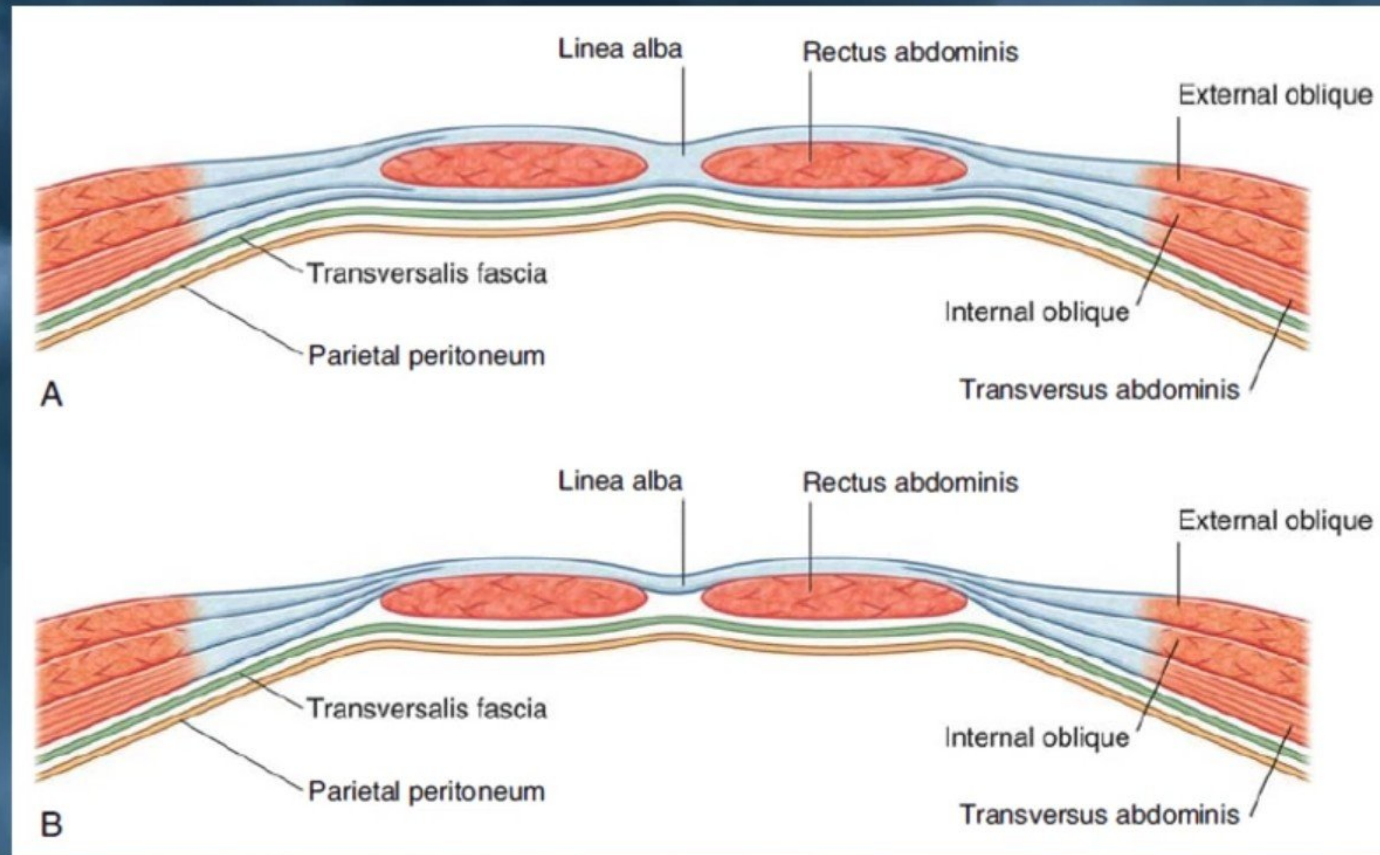
Arrangement of the rectus sheath

- A) Transverse section through **the upper three quarters of the rectus sheath** (above the umbilicus).
- B) Transverse section thorough the **lower one quarter of the rectus sheath** (below the arcuate line).



Rectus sheath

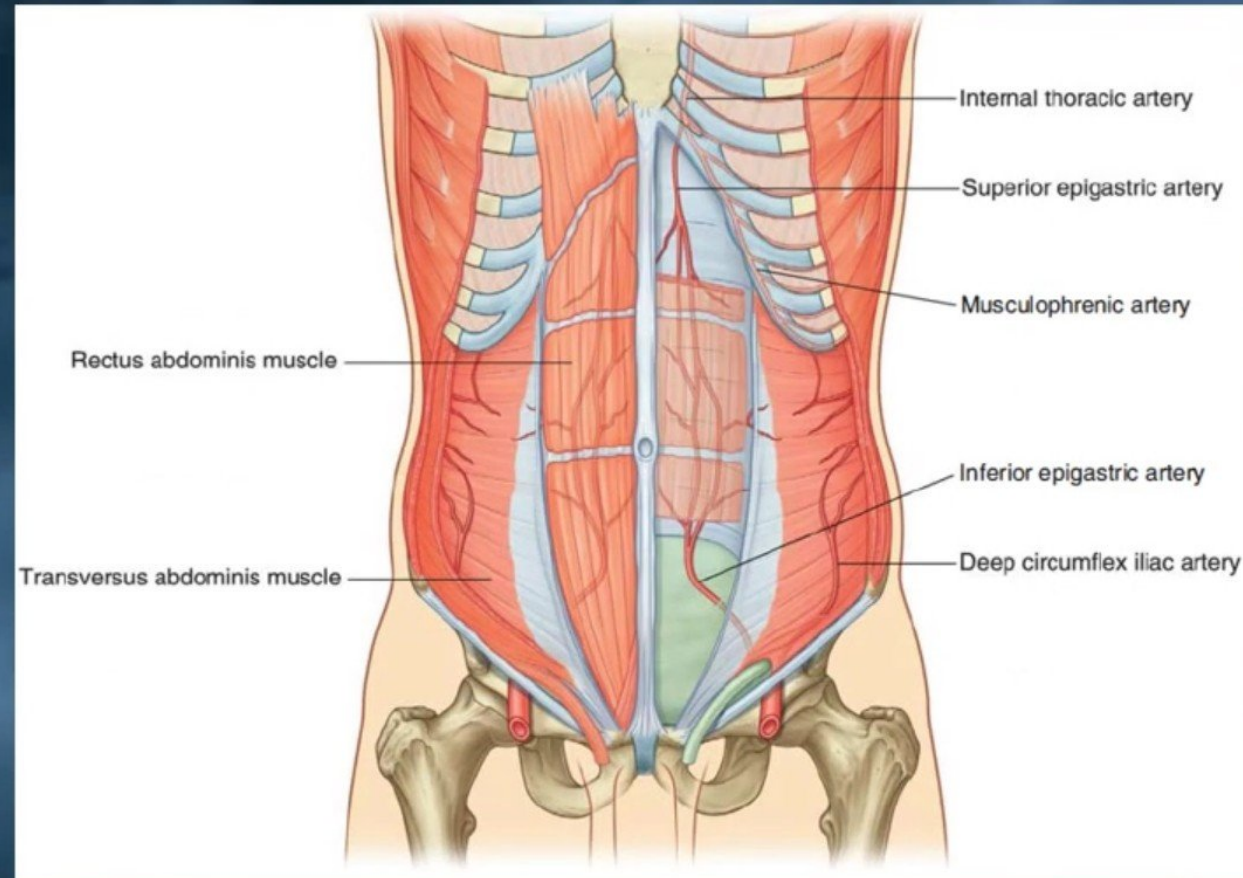
- The aponeurosis of the internal oblique **splits** into **anterior** and **posterior layers** to enclose the rectus muscle .
- The **external oblique aponeurosis fuses** with the **anterior layer** to form the **anterior layer of the sheath**, and the **transversus aponeurosis fuses** with the **posterior layer** to form the **posterior layer of the sheath**.
- From halfway between the umbilicus and the pubic symphysis all three aponeuroses pass in front of the muscle.



Contents of the Rectus sheath:

- 1) The rectus abdominis muscle.
- 2) Pyramidalis muscle.
- 3) The ends of the lower six thoracic nerves (intercostal nerve).
- 4) The lower posterior six intercostal vessels.
- 5) The superior and inferior epigastric vessels.

Rectus sheath

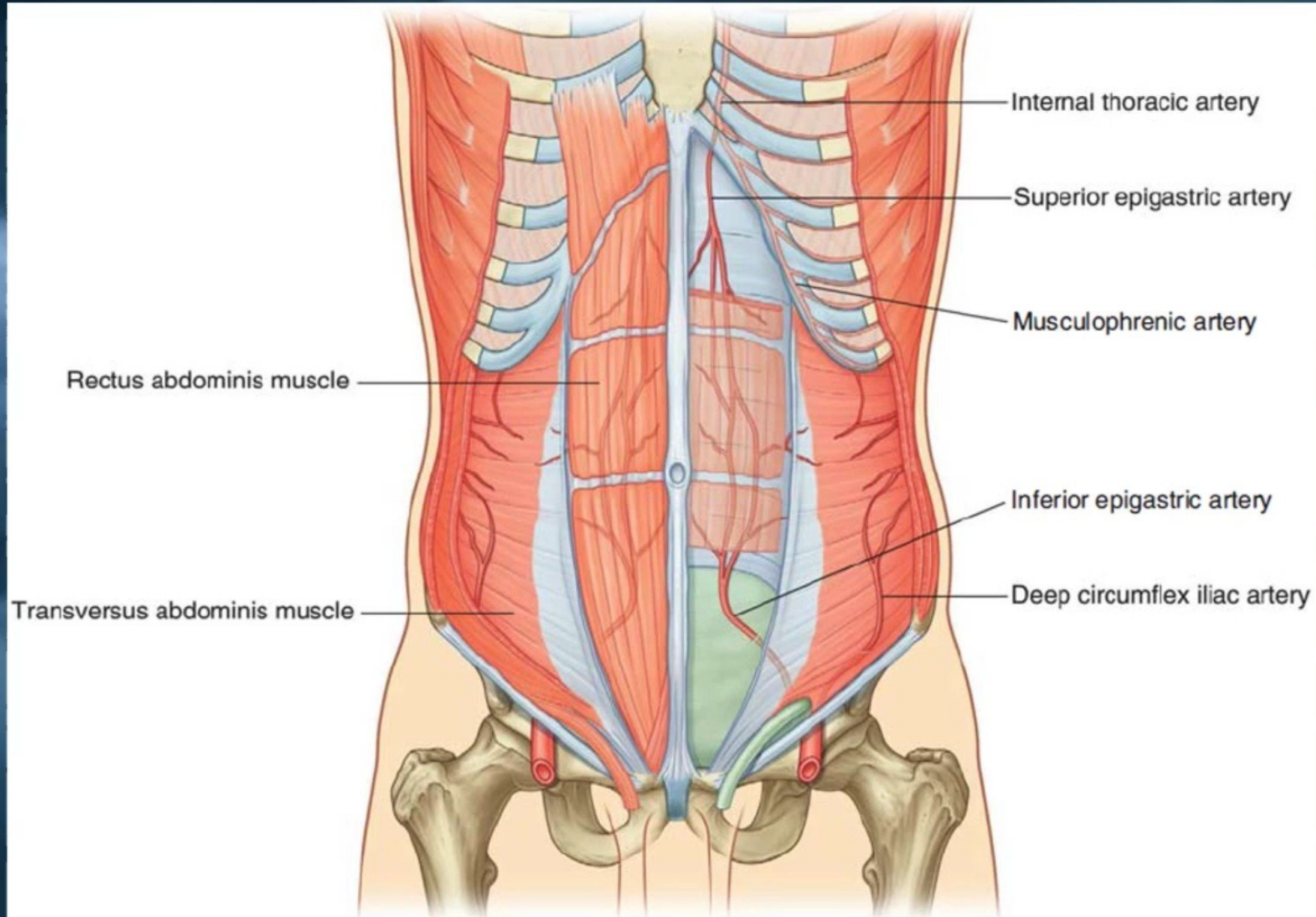


Rectus sheath

The vessels found in rectus sheath.

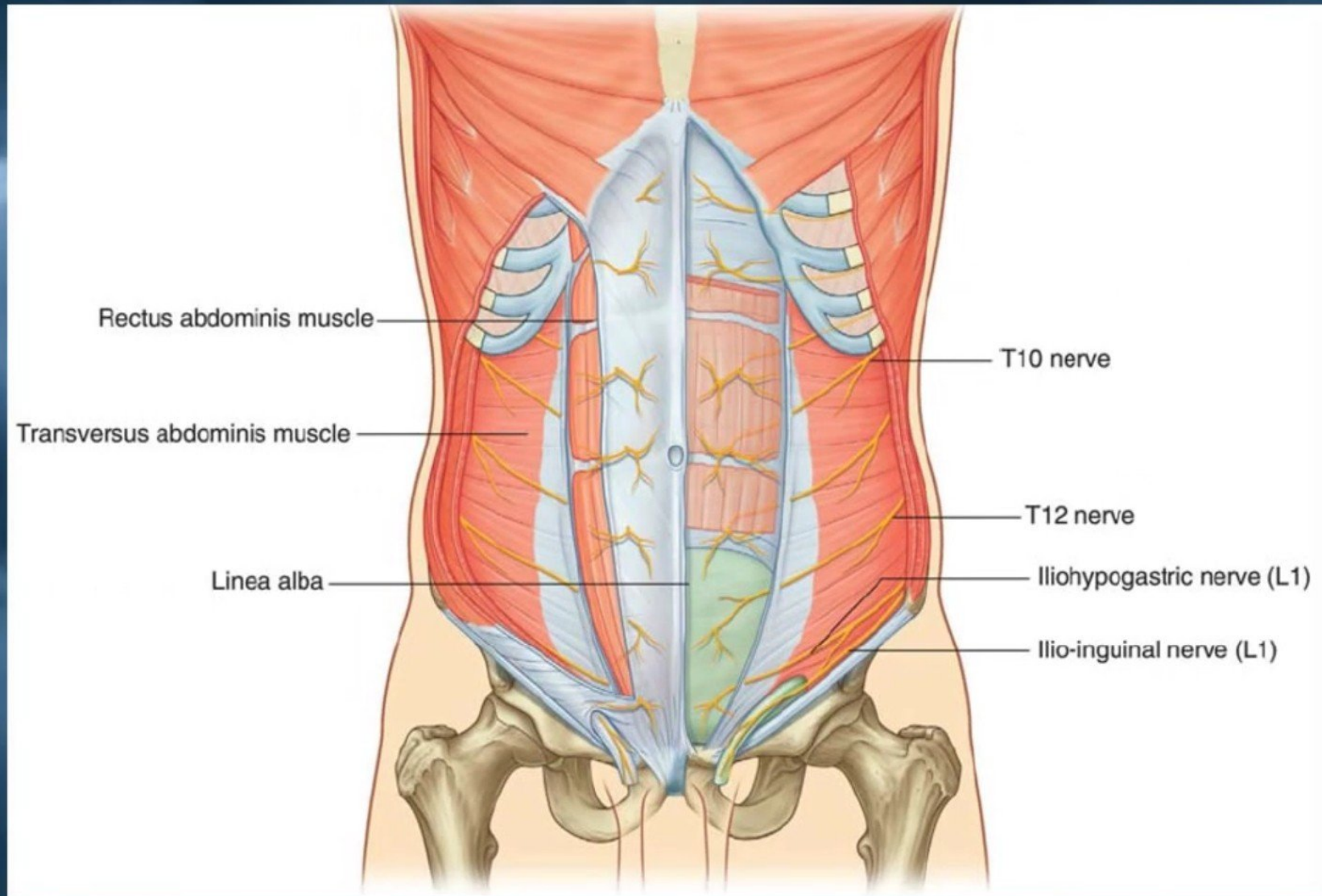
The lower posterior six intercostal vessels.

The superior and inferior epigastric vessels



Rectus sheath

- The ends of the **lower six thoracic nerves** (intercostal nerve).
- The **iliohypogastric.**
- **ilioinguinal nerves (L1).**



Muscle	Origin	Insertion	Nerve	Action
External oblique	External surface of lower eight ribs (5–12)	Anterior half of iliac crest; anterior–superior iliac spine; pubic tubercle; linea alba	Intercostal (T7–T11); subcostal (T12)	Compresses abdomen; flexes trunk; active in forced expiration
Internal oblique	Lateral two-thirds of inguinal ligament; iliac crest; thoracolumbar fascia	Lower four costal cartilages; linea alba; pubic crest; pectineal line	Intercostal (T7–T11); subcostal (T12); iliohypogastric and ilioinguinal (L1)	Compresses abdomen; flexes trunk; active in forced expiration
Transverse	Lateral one-third of inguinal ligament; iliac crest; thoracolumbar fascia; lower six costal cartilages	Linea alba; pubic crest; pectineal line	Intercostal (T7–T12); subcostal (T12); iliohypogastric and ilioinguinal (L1)	Compresses abdomen; depresses ribs
Rectus abdominis	Pubic crest and pubic symphysis	Xiphoid process and costal cartilages fifth to seventh	Intercostal (T7–T11); subcostal (T12)	Depresses ribs; flexes trunk
Pyramidal	Pubic body	Linea alba	Subcostal (T12)	Tenses linea alba
Cremaster	Middle of inguinal ligament; lower margin of internal oblique muscle	Pubic tubercle and crest	Genitofemoral	Retracts testis

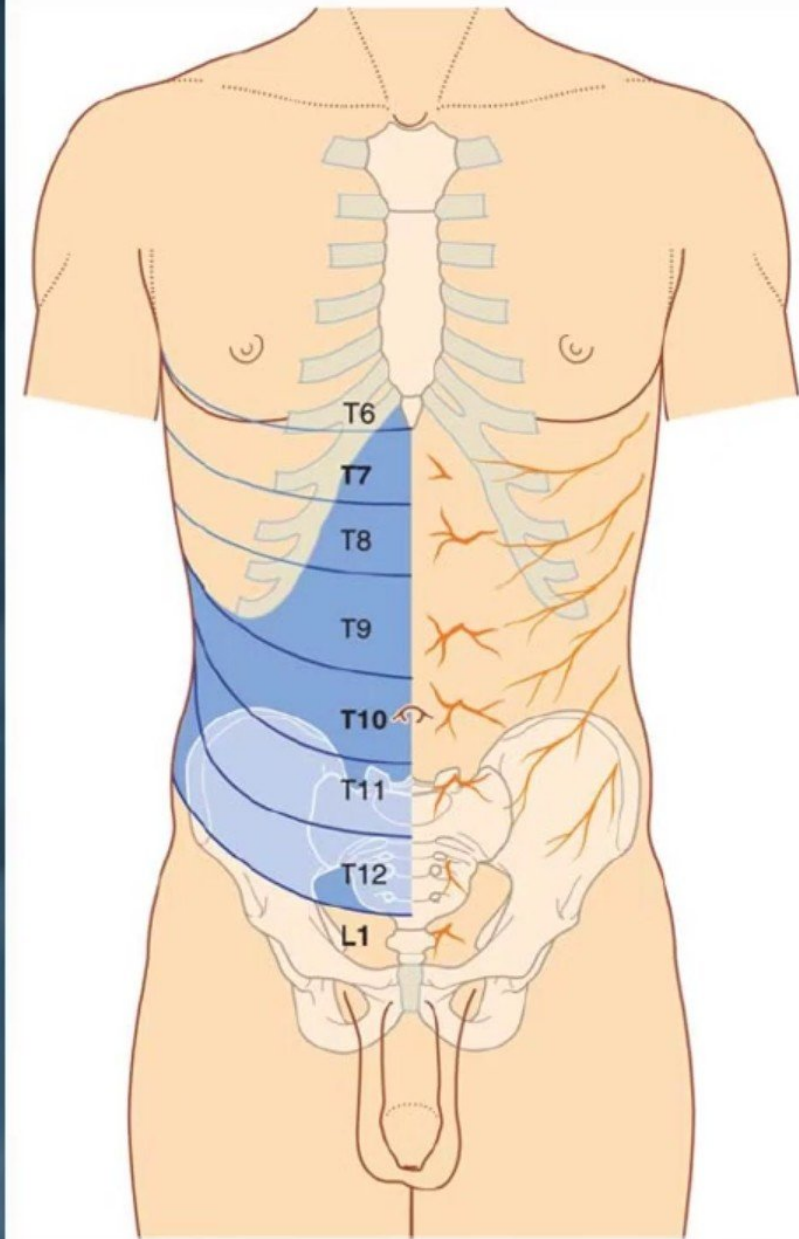
The inguinal ligament

- The lower border, lying between **the anterior superior iliac spine** and **the pubic tubercle**, forms **the inguinal ligament (of Poupart)**.
- **The lateral part of inguinal ligament** gives origin to part of **the internal oblique** and **transversus abdominis muscles**.
- **The fascia lata** of the thigh is attached to **the inguinal ligament** and when the thigh is extended the fascia lata pulls the inguinal ligament downwards into a gentle convexity.
- Just **above** and **lateral** to **the pubic tubercle** is an oblique, **triangular gap, the superficial inguinal ring**, in the aponeurosis.

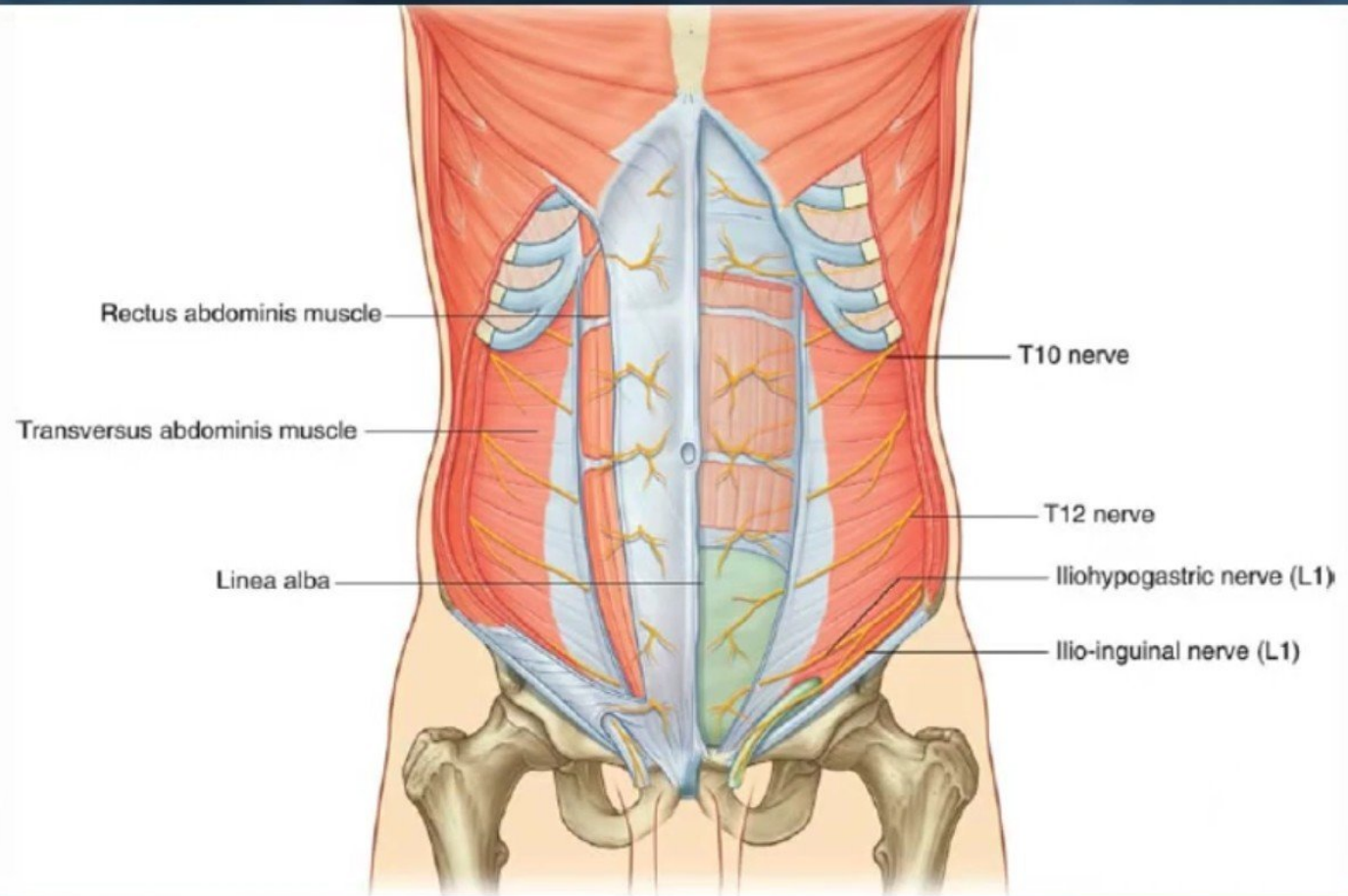
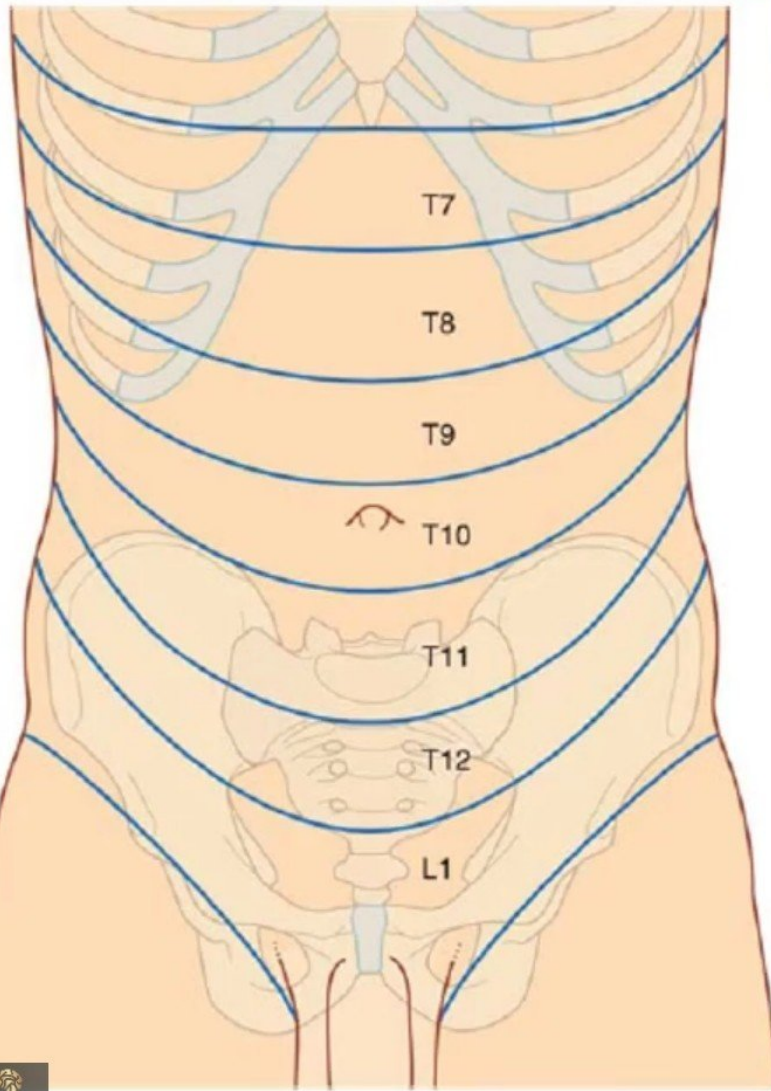
Cutaneous innervation Of the anterior abdominal wall

Skin innervation:

- lower 5 intercostal nerves.
- Subcostal nerve.
- L1 spinal nerve (ilioinguinal + iliohypogastric nerves).
- Umbilical region skin innervation of T10.



Innervation Of the anterior abdominal wall



Blood vessels of the anterior abdominal wall

A. Superior Epigastric artery.

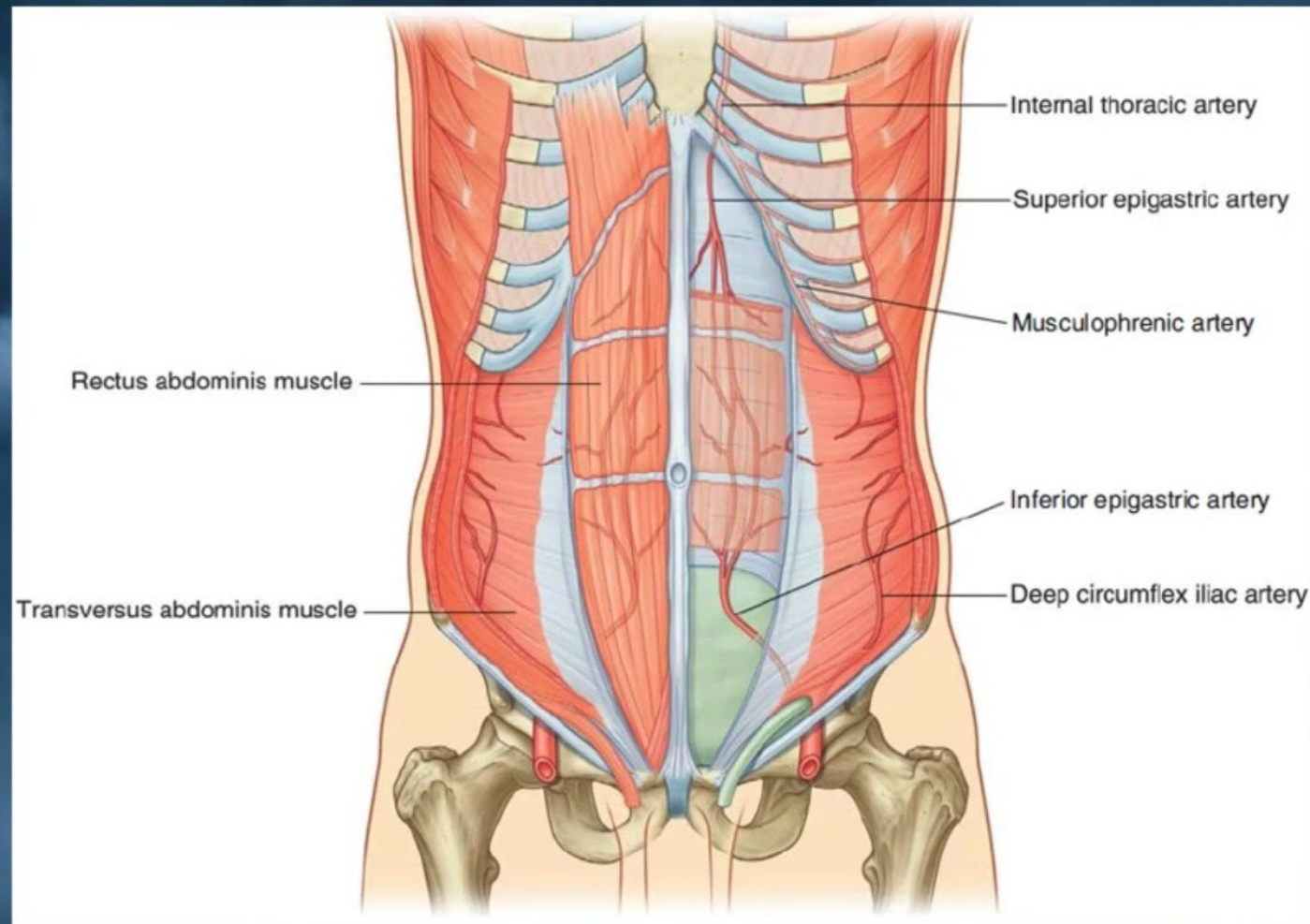
- It arises from the internal thoracic artery.

B. Inferior Epigastric artery.

- It arises from the external iliac artery.

C. Deep Circumflex Iliac artery.

- It arises from the external iliac artery .



Blood vessels of the anterior abdominal wall

D. Superficial Epigastric arteries.

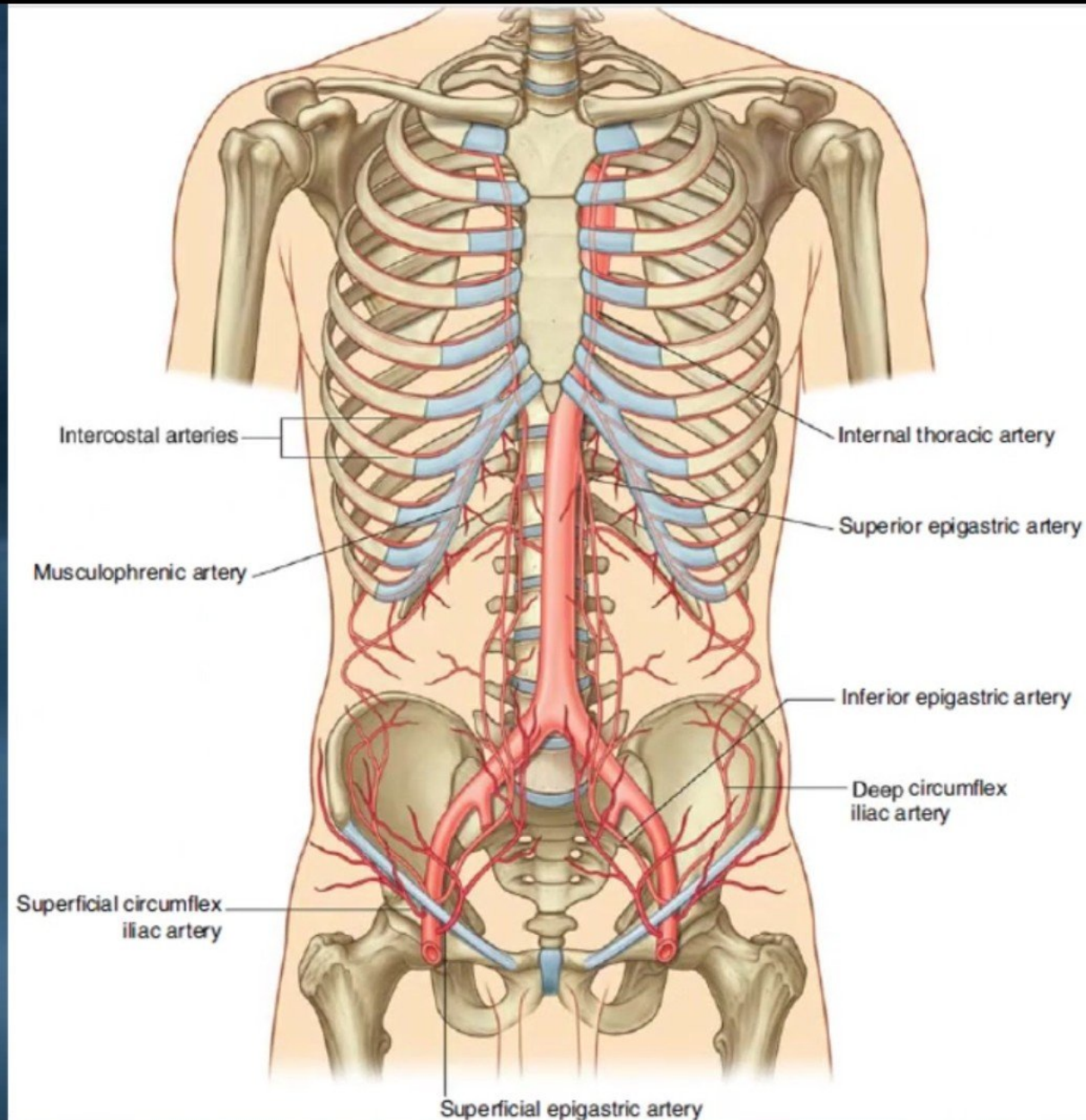
- It arise from the femoral artery.

E. Superficial Circumflex iliac artery

- It arises from the femoral artery.

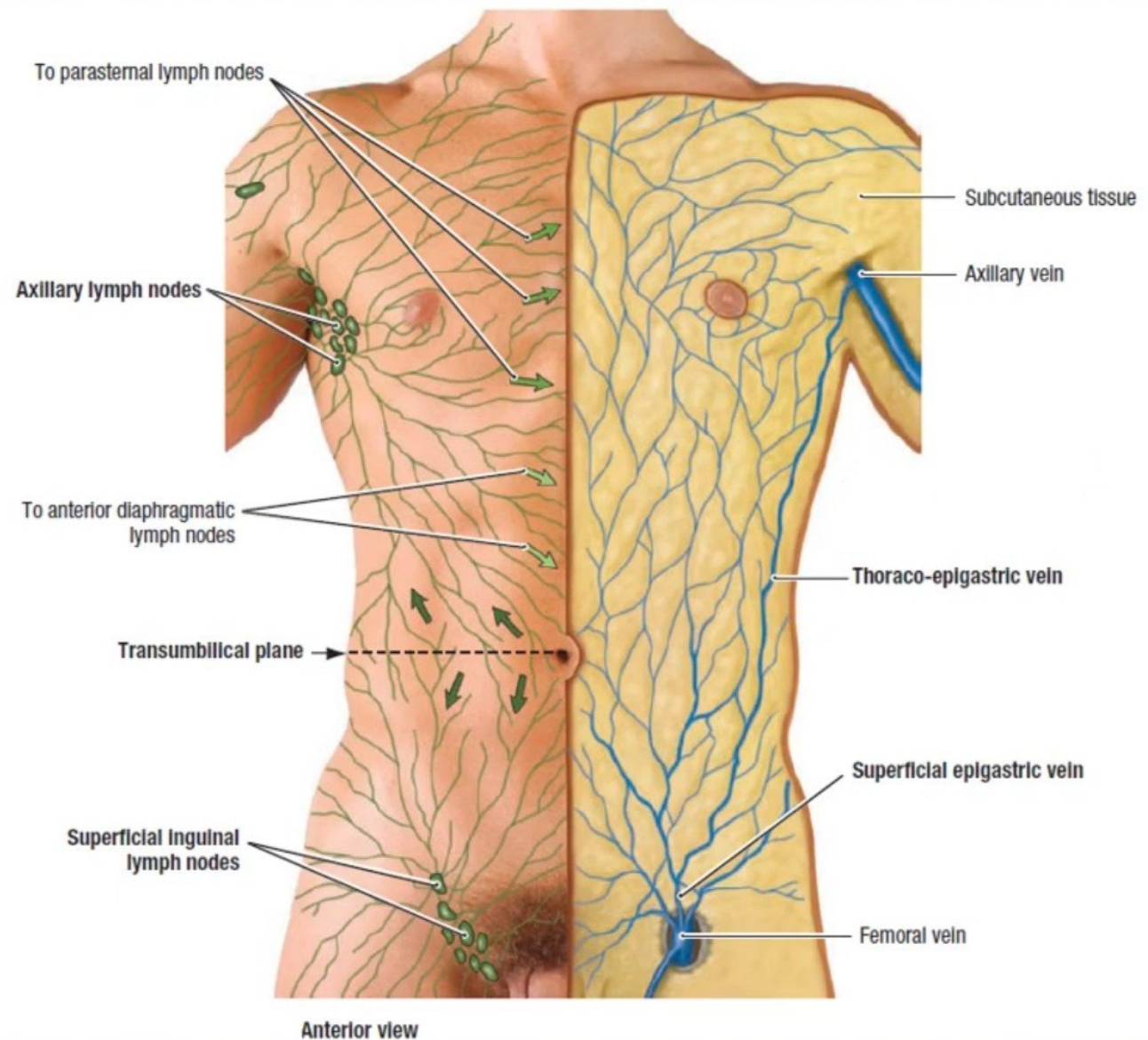
F. Superficial (External) pudendal arteries

- It arise from the femoral artery.



Venous Drainage

- Venous blood is collected into a **network of veins** that **radiate** from **the umbilicus**.
- The network is drained **above** into **the axillary vein** via the **lateral thoracic vein**.
- **Below** into **the femoral vein** via the **superficial epigastric** and the **great saphenous veins**.
- Few small veins, the **paraumbilical veins** form a clinically important **portal-system venous anastomosis**.



Lymphatic Drainage

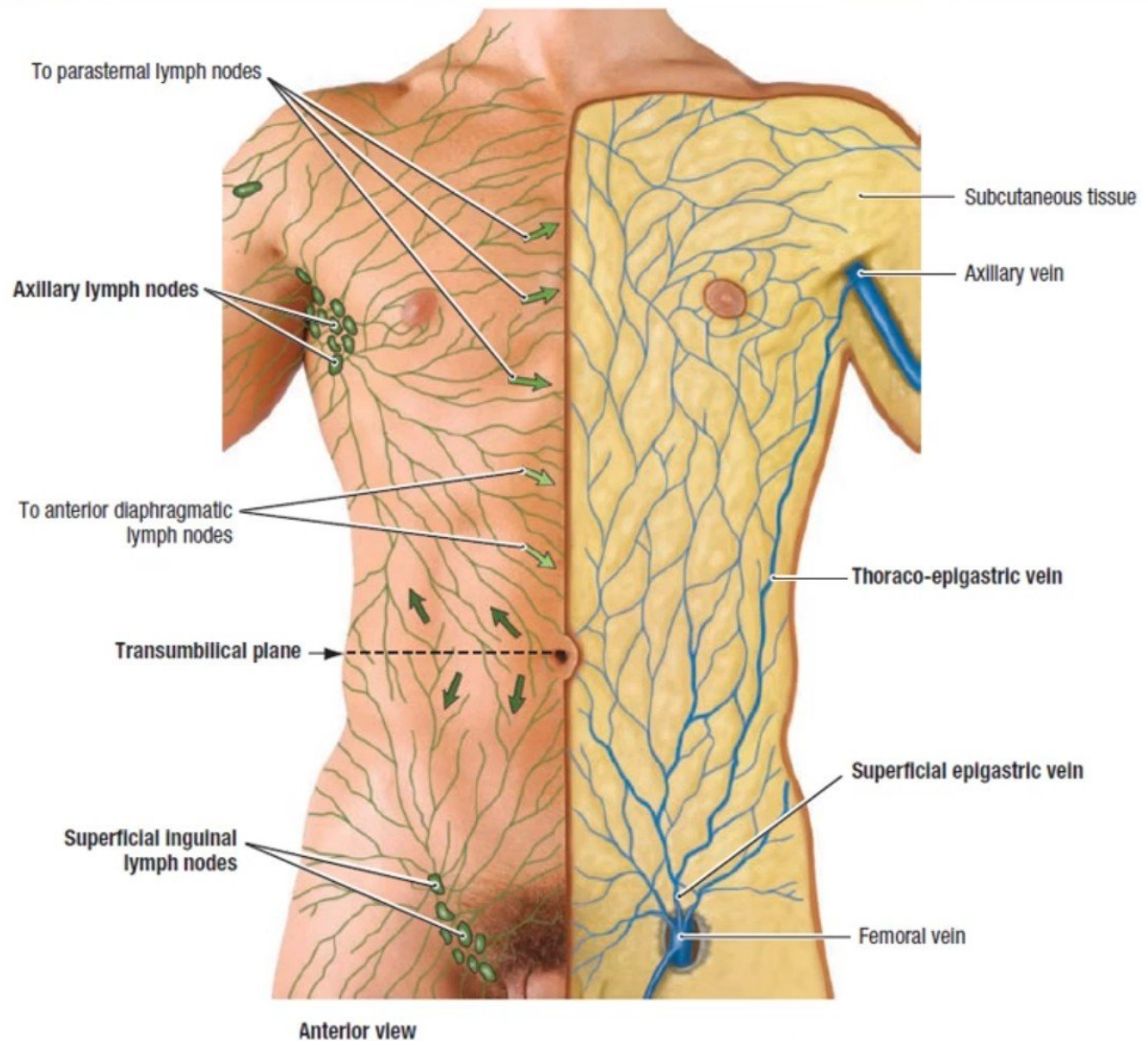
Superficial
Inguinal lymph
nodes

Axillary
Lymph nodes

Lymph node:

Axillary - above the umbilical
region.

Inguinal - below the umbilical
region.



Thanks

